

EE104 Fundamentals of Electric Circuits (Lesson 03)

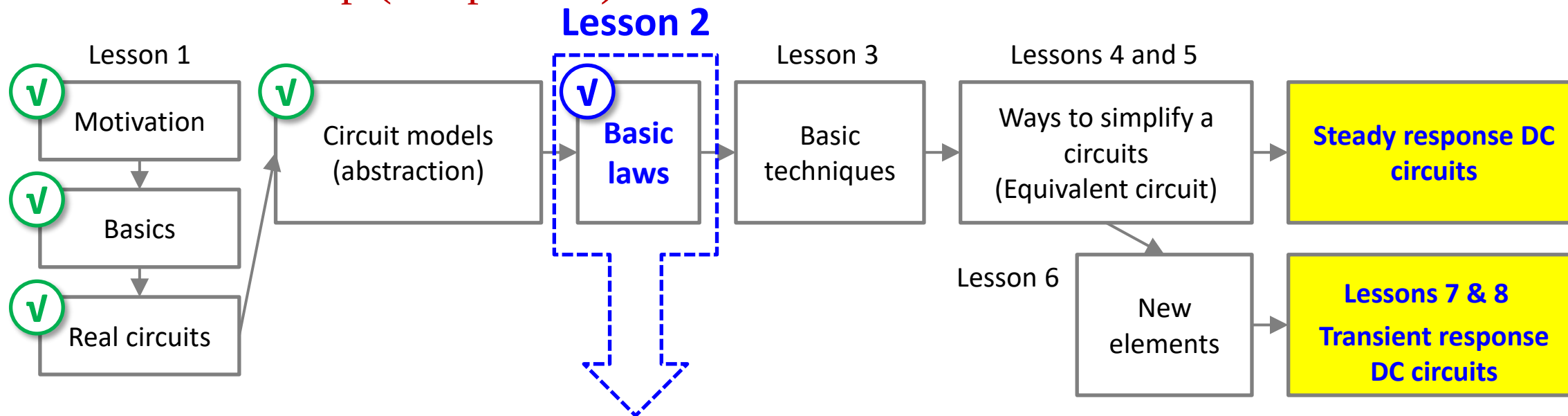
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Review of Lesson 2 (a)

- EE104 roadmap (DC portion)



- a) The circuits big three
- Ohm's law
 - KCL
 - KVL
- b) Equivalent circuits
- R in series/in parallel (current/voltage division)
 - Y- Δ networks
 - Etc.

Review of Lesson 2 (b)

- THE CIRCUITS BIG THREE, DO NOT FORGET !!!

Ohm's law

Ohm's law states that the voltage v across a conductor is directly proportional to the current i flowing through it, provided all physical conditions and temperature remain constant.

KCL

The algebraic sum of currents entering a node (or a closed boundary) is zero. In other words, the sum of the currents entering a node is equal to that leaving the node.

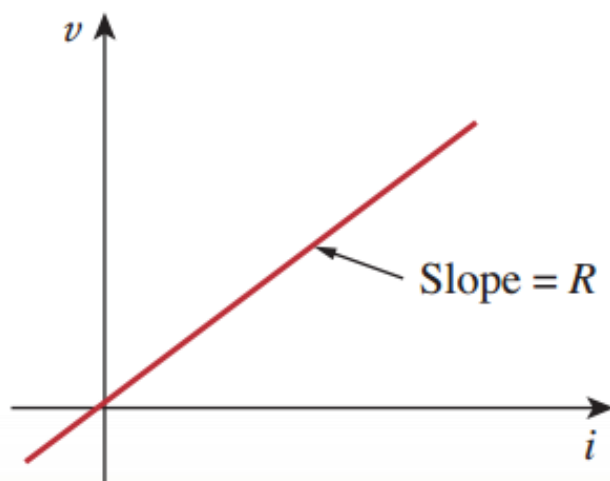
KVL

The algebraic sum of all voltages around a loop is zero, or Sum of voltage drops = Sum of voltage rises, which is based on the conservation of energy.

Review of Lesson 2 (c)

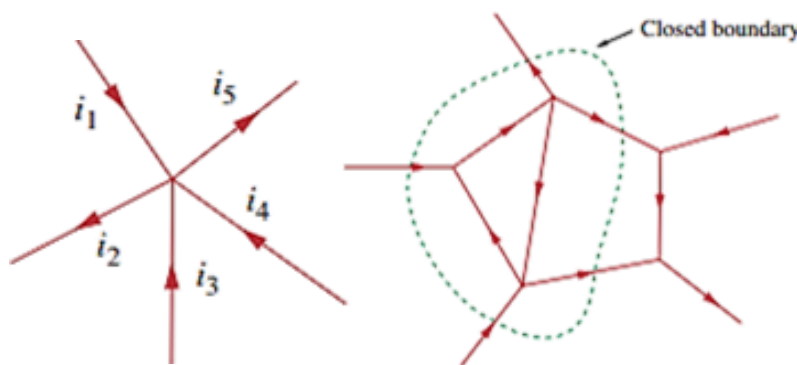
- The circuits big three !!!

Ohm's law



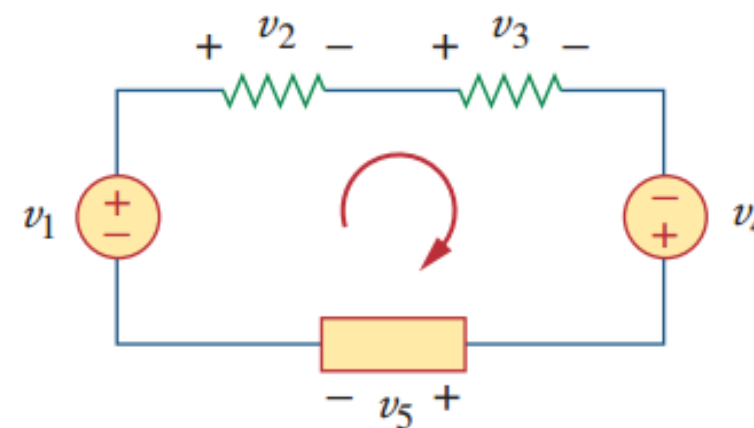
$$v = iR \text{ or } v = -iR$$

KCL



- 1) the conservation of charge
- 2) applies to nodes and closed boundaries
- 3) is based on reference current directions
- 4) $+i$ when in and $-i$ when out

KVL



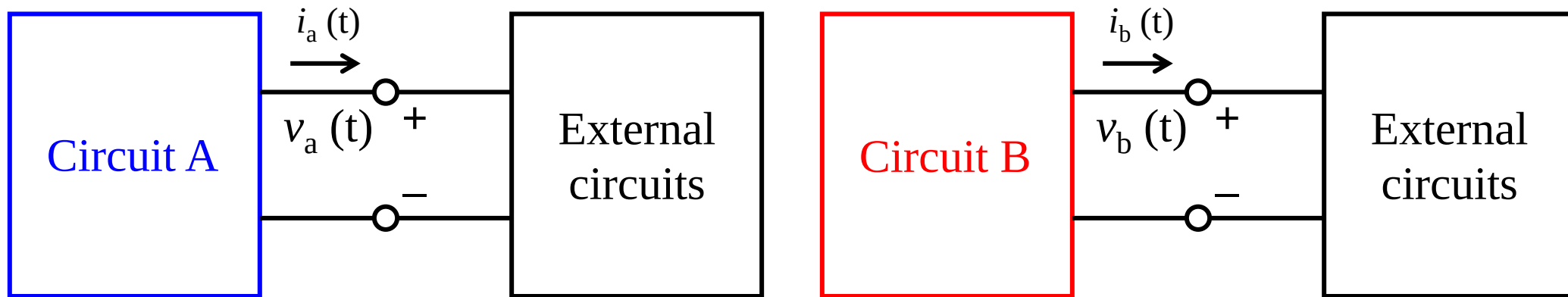
- 1) the conservation of energy
- 2) applies to loops and meshes
- 3) is based on reference voltage directions
- 4) $+v$ when associated and $-v$ when non associated

Review of Lesson 2 (d)

- Equivalent circuits:

Circuits A and B are equivalent to external circuits when they yield the same i - v characteristics at their ports. At the port, the circuit's internal details, its design, and its theory of operation are no longer important. It becomes a building block or a black box.

- a) The same i - v characteristics at the ports.
- b) Equivalent circuits are equivalent as to external circuits, not internal.

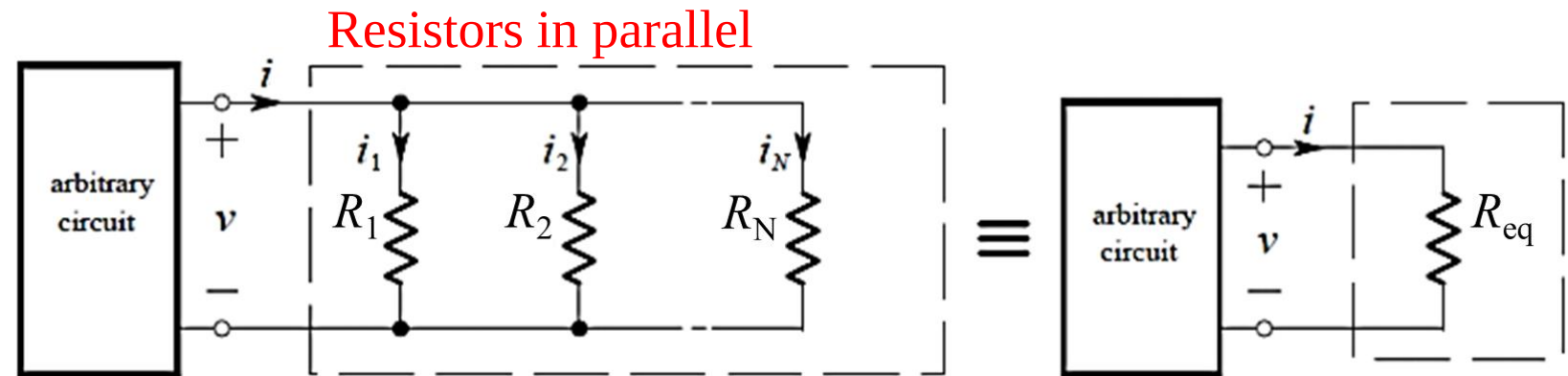
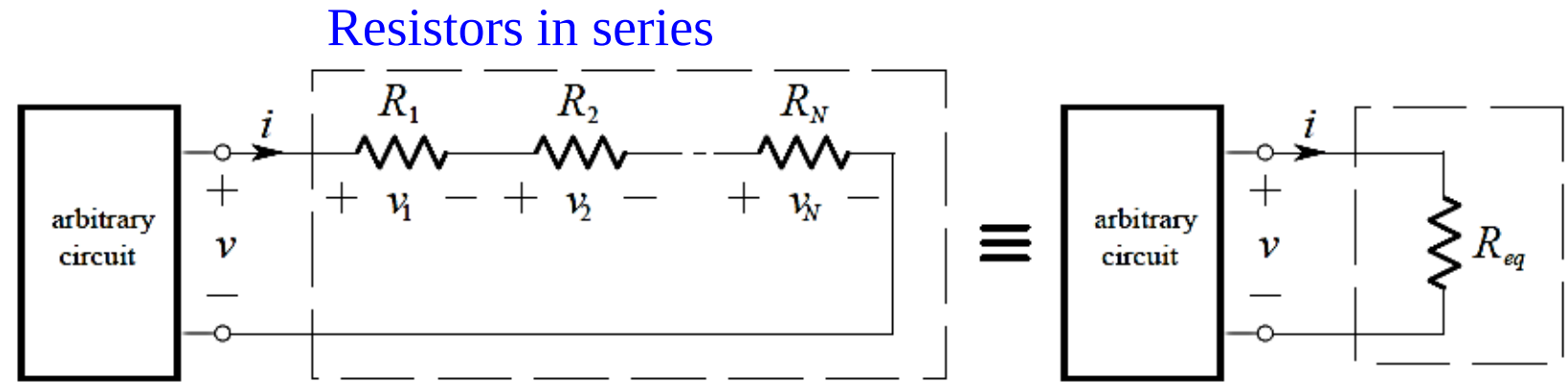


Always remember how to make two circuits equivalent!

Review of Lesson 2 (e)

- Equivalent circuits

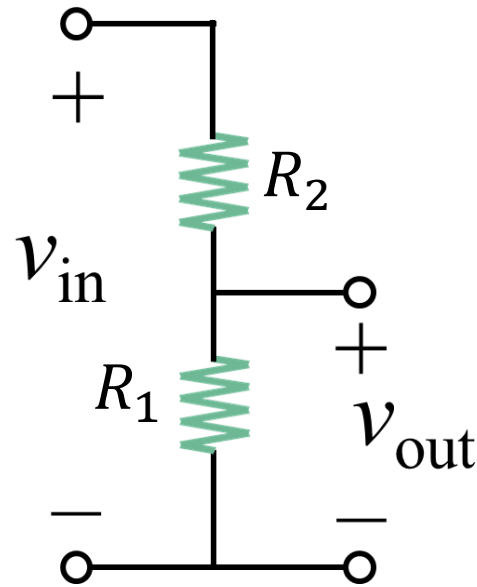
- a) An **equivalent circuit** is a simplified model of an existing circuit, used to simplify the analysis.
- b) Two circuits are said to be equivalent if they have **identical v - i characteristics** at the ports.
- c) It is **of vast importance** to understand the concept of equivalent circuits.



Review of Lesson 2 (f)

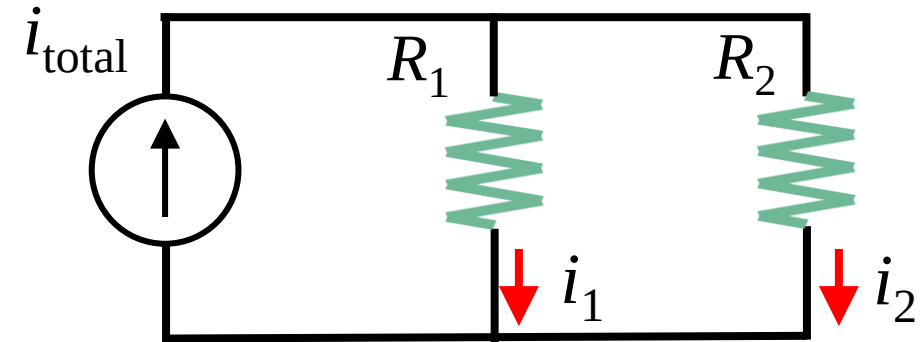
- Voltage division and current division

Voltage
division



$$R_{eq} = R_1 + R_2 \quad v_{out} = \frac{R_1}{R_1 + R_2} v_{in}$$

Current division



$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} \quad G_{eq} = G_1 + G_2$$

$$i_1 = \frac{G_1}{G_1 + G_2} i_{total}$$

Review of Lesson 2 (g)

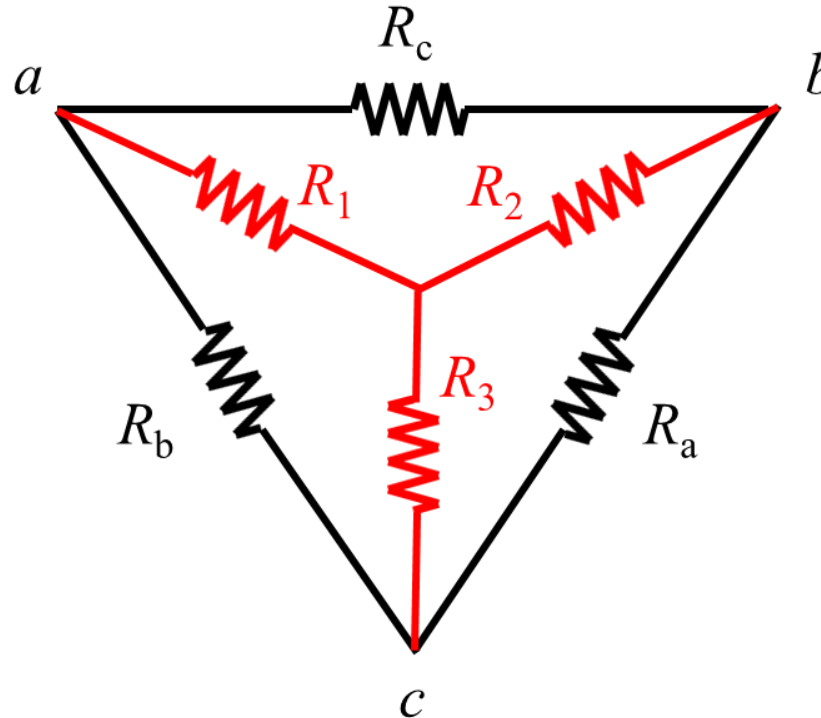
- Y- Δ and Δ -Y transformations

Δ to Y

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_a R_c}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_b R_a}{R_a + R_b + R_c}$$



Y to Δ

$$R_a = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_1}$$

$$R_b = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_2}$$

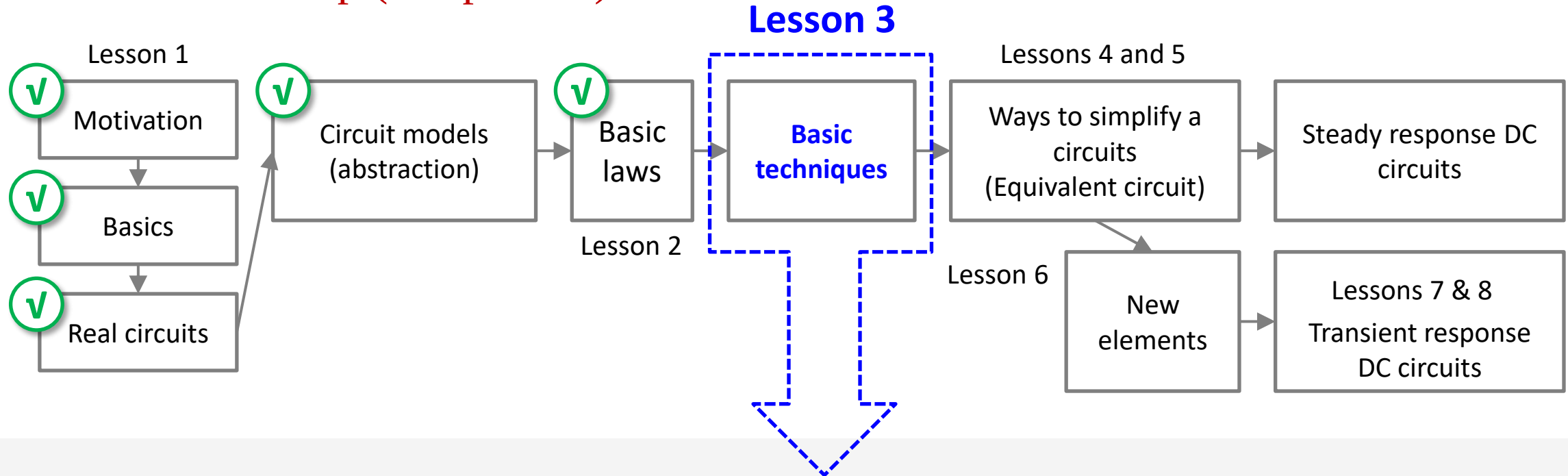
$$R_c = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_3}$$

Δ to Y: the resistance of a resistor in the Y network is the product of the resistance of the two adjacent resistors in the Δ network, divided by the sum of the resistance of the three Δ resistors.

Y to Δ : The resistance of a resistor in the Δ network is the sum of all possible products the resistance of the Y resistors taken two at a time, divided by the resistance of the opposite Y resistor.

Review of Lesson 2 (h)

- EE104 roadmap (DC portion)



Nodal analysis & Mesh analysis

(planar circuits & supernode & supermesh)

Lesson 3 Systematic approaches for circuit analysis

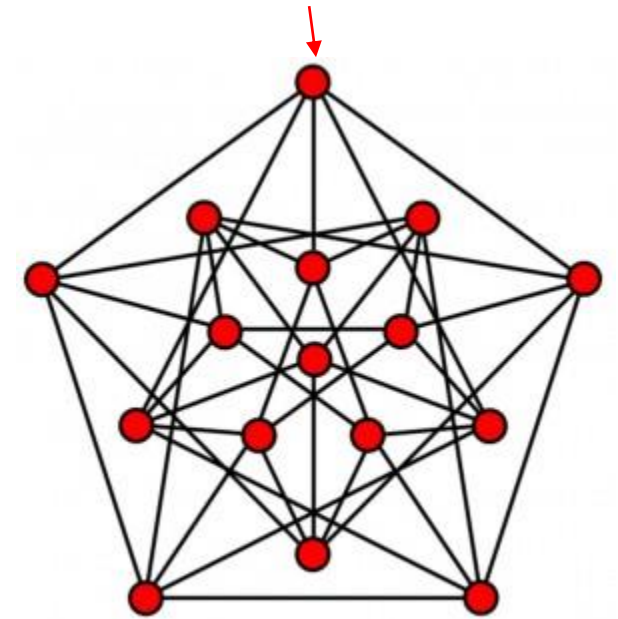
3.1-3.2	Graph theory	Slides 11-19
3.3-3.7	Nodal voltage analysis	Slides 21-52
3.8-3.9	Mesh current analysis	Slides 54-67
3.10	Comparison of Nodal and mesh analysis	Slides 79-75
3.11	Application: DC transistor circuits	Slides 77-82
3.12	Summary and assignments	Slides 84-86

3.1 Graph theory (a)

- Graph and graph theory :

- a) A graph is a diagram of points and lines connected to the points. It has at least one line joining a set of two vertices with no vertex connecting itself.
- b) A **vertex** is a point where multiple lines meet. It is also called a node.
- c) An **edge** is the mathematical term for a line that connects two vertices.
- d) In mathematics, **graph theory** (图论) is the study of graphs, which are mathematical structures used to model pairwise relations between objects. It is a discrete mathematics sub-branch and is at the highest level the study of connection between things.

vertices (points or nodes)

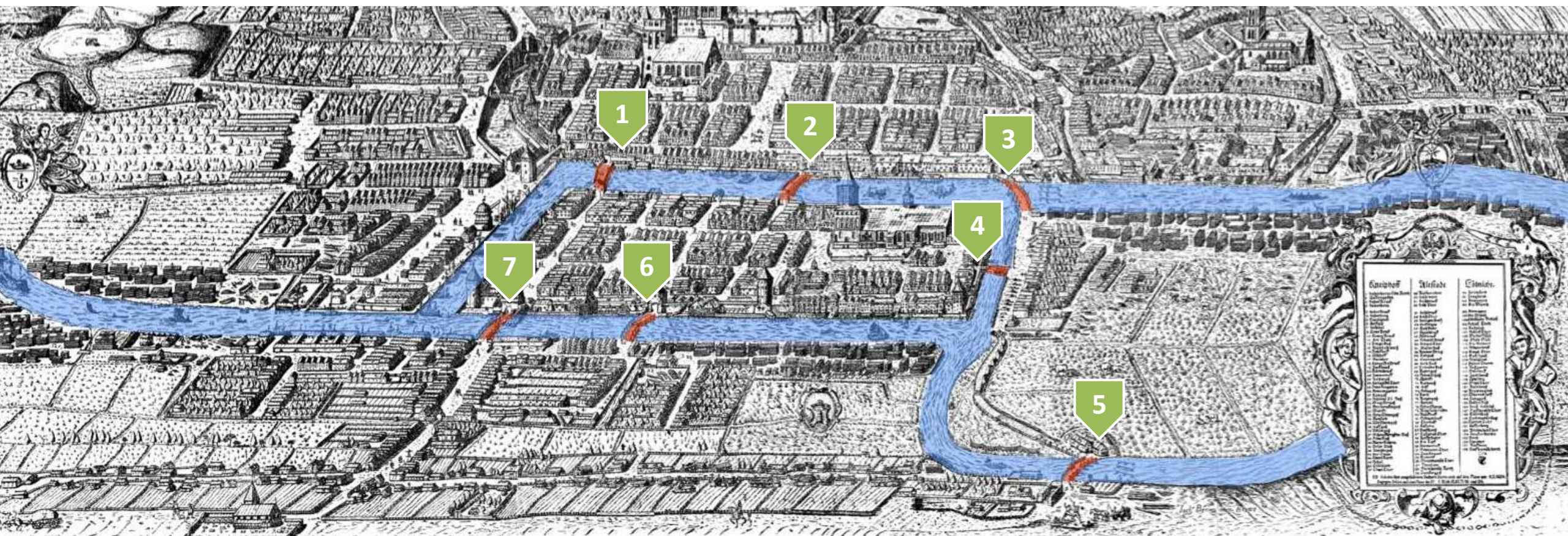


edges (or lines)

3.1 Graph theory (b)

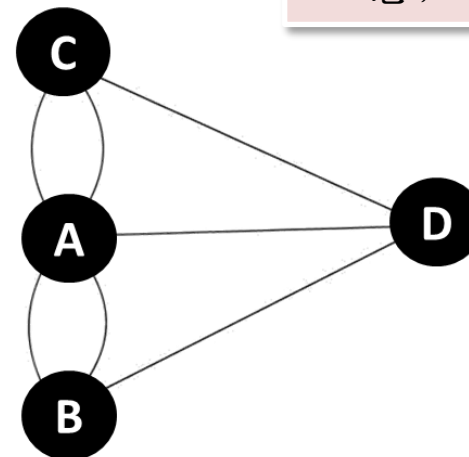
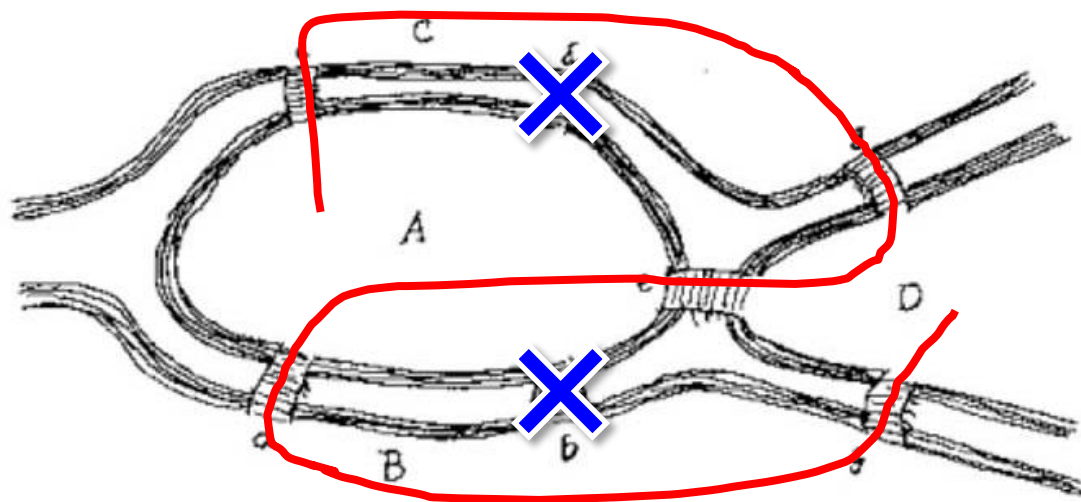
- **Seven bridges of Königsberg (哥尼斯堡七桥):**

Can we walk around the city in a way that would involve crossing each bridge exactly once?



3.1 Graph theory (c)

- Euler and graph theory:



还需一座桥，
急，在线等

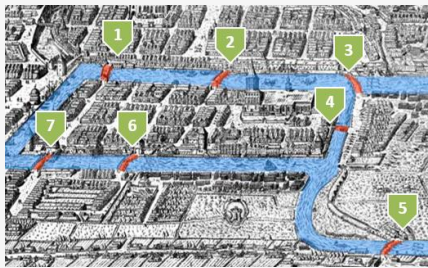


- The first known work on graph theory was Leonhard Euler's (欧拉) paper on The Seven bridges of Königsberg in 1736.
- Leonhard Euler was a Swiss mathematician, physicist, astronomer, geographer, logician and engineer.
- Can we traverse each bridge of Königsberg once and only once? No

3.1 Graph theory (d)

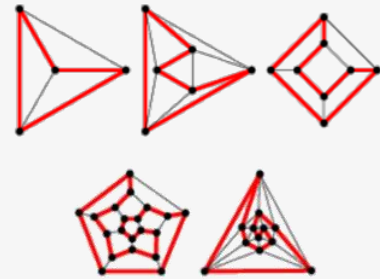
- History of graph theory:

“Seven Bridges of Königsberg”



Leonhard Euler

Cycles in Polyhedra

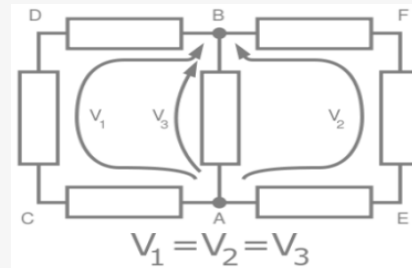


Thomas P. Kirkman



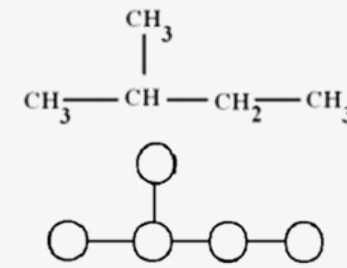
William R. Hamilton

Trees in Electric Circuits



Gustav Kirchhoff

Enumeration of Chemical Isomers



Arthur Cayley

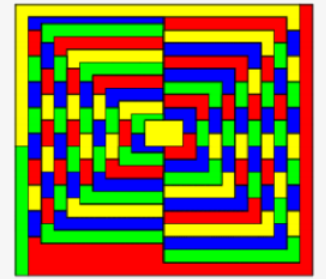


James J. Sylvester



George Polya

Four Colors of Maps



Francis Guthrie



Auguste DeMorgan

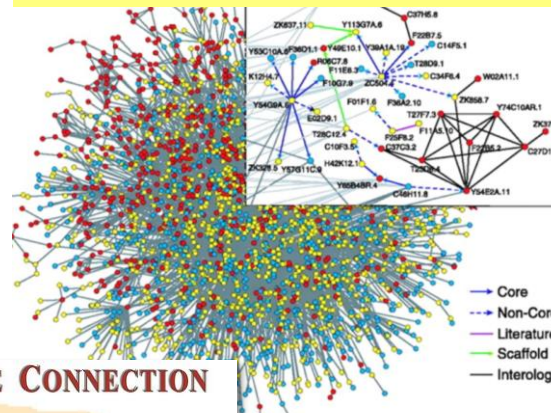
3.1 Graph theory (e)

- Application of graph theory:

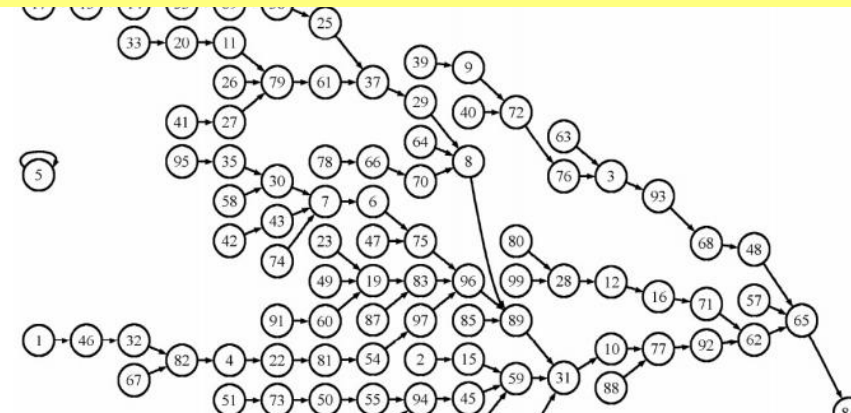
Social networks



Network biology

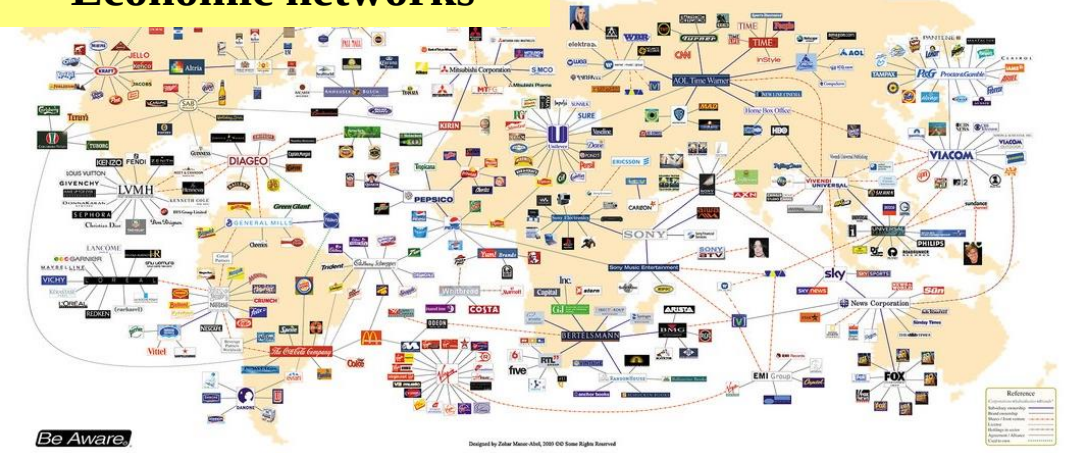


Graph theory and cryptography



Economic networks

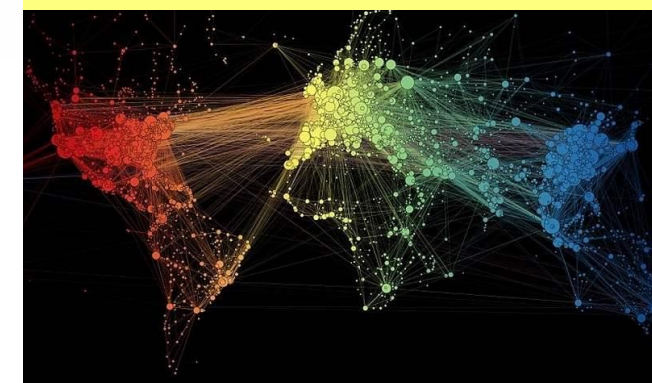
CORPORATE CONNECTION



Information networks



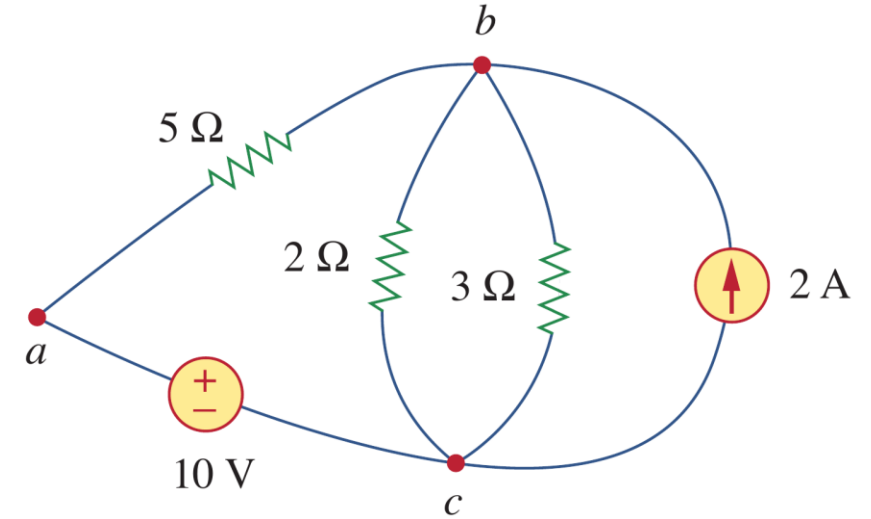
Traffic networks



3.2 Network topology (a)

- Graph theory and electric circuits:

- a) Graph theory plays very crucial role in understanding of complicated electric circuits. It is very useful for analyzing complicated circuits.
- b) A **branch** represents a single element such as a voltage source or a resistor.
- c) A **node** is the point of connection between two or more branches.
- d) A **loop** is a closed path in a circuit.
- e) A **mesh** is a loop which does not contain any other loops within it.



5 branches, $b = 5$

3 nodes, $n = 3$

3 meshes, $m = 3$

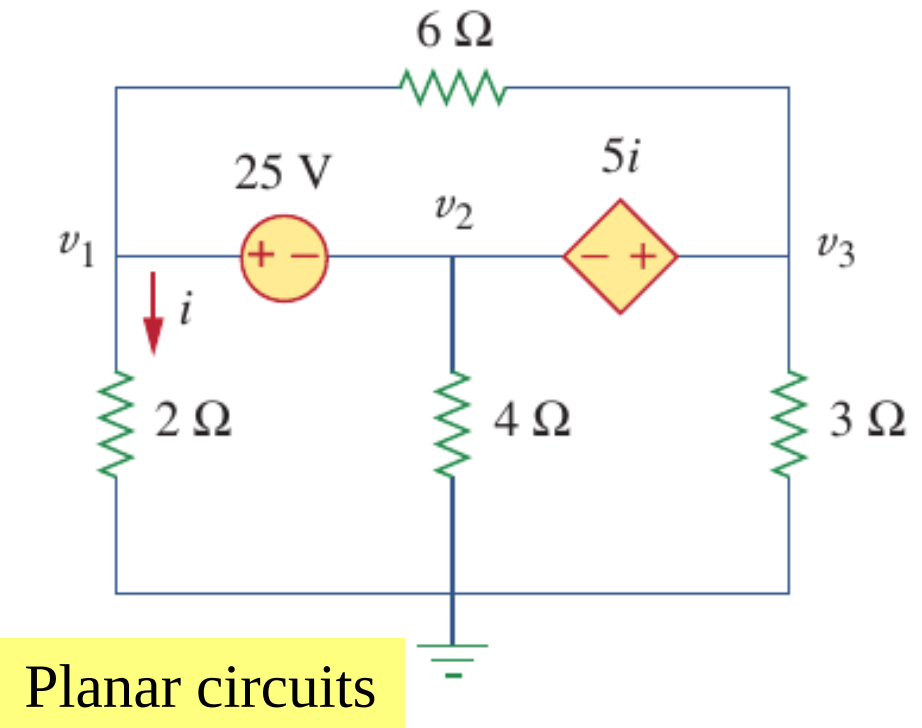
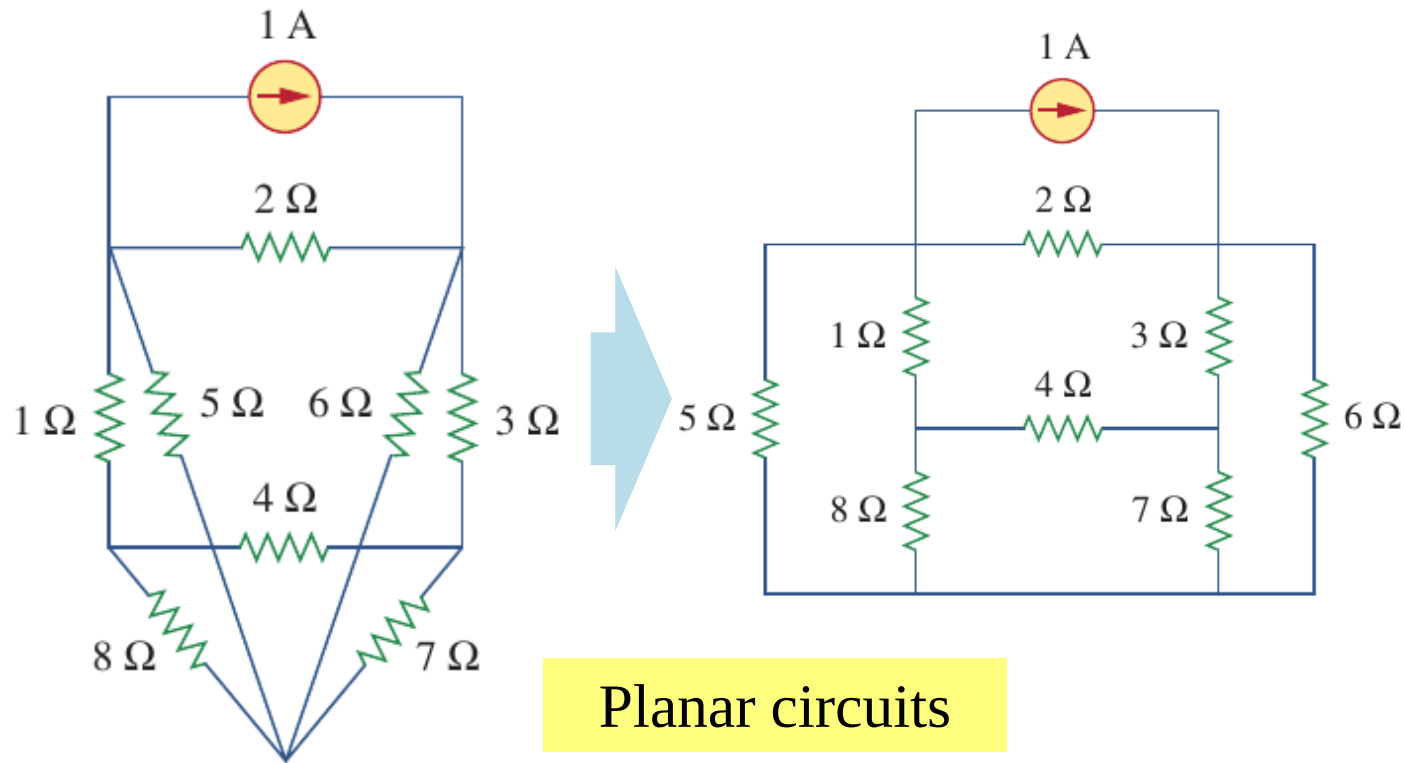
6 loops

$$b = m + n - 1$$

3.2 Network topology (b)

- Planar circuits:

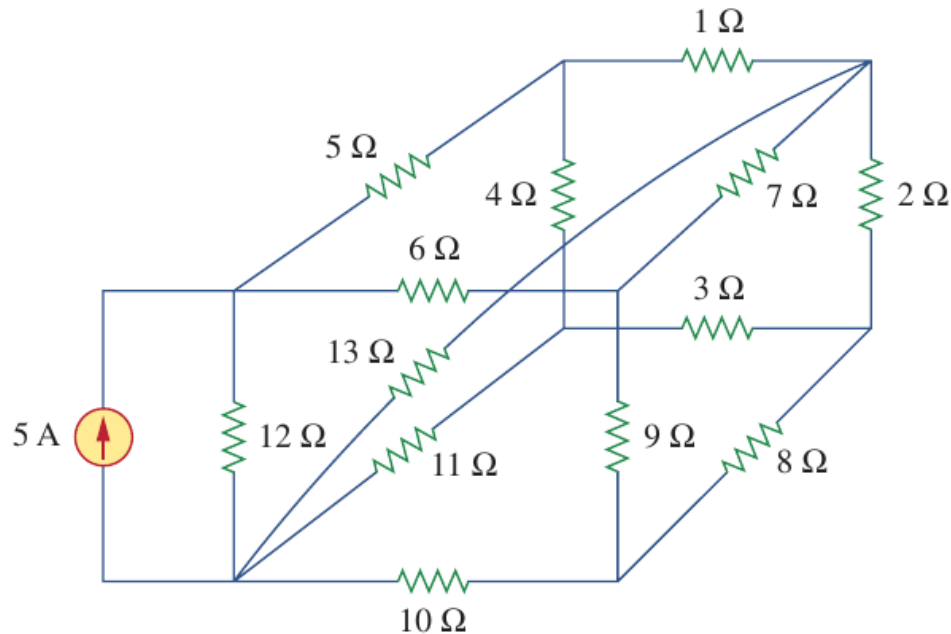
□ A planar circuit is one that can be drawn in a plane with no branches crossing one another; otherwise it is nonplanar.



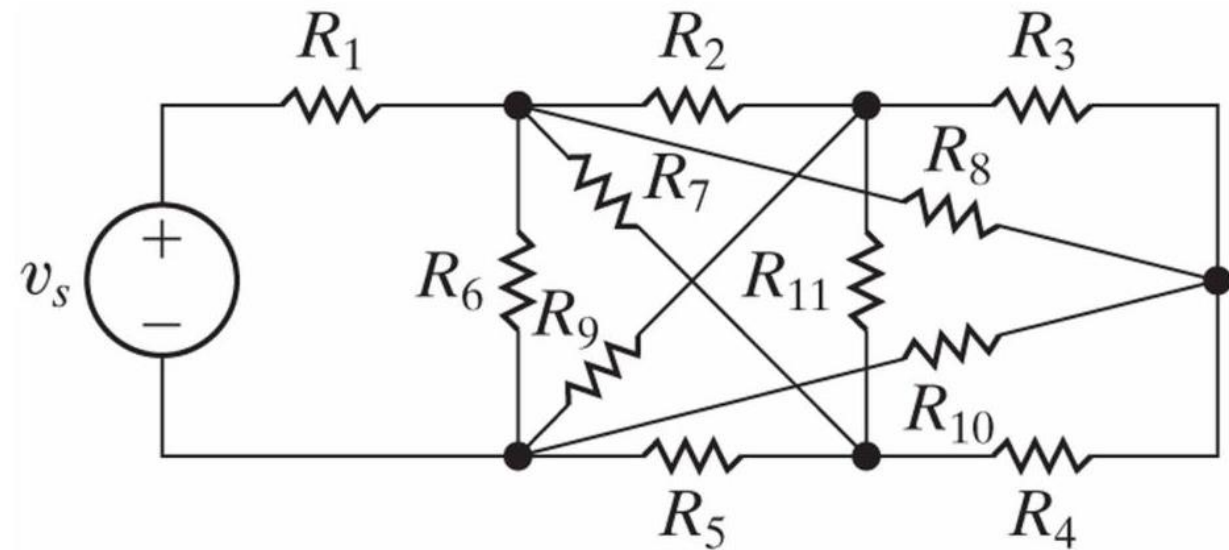
3.2 Network topology (c)

- Non-planar circuits:

□ A planar circuit is one that can be drawn in a plane with no branches crossing one another; otherwise it is nonplanar.

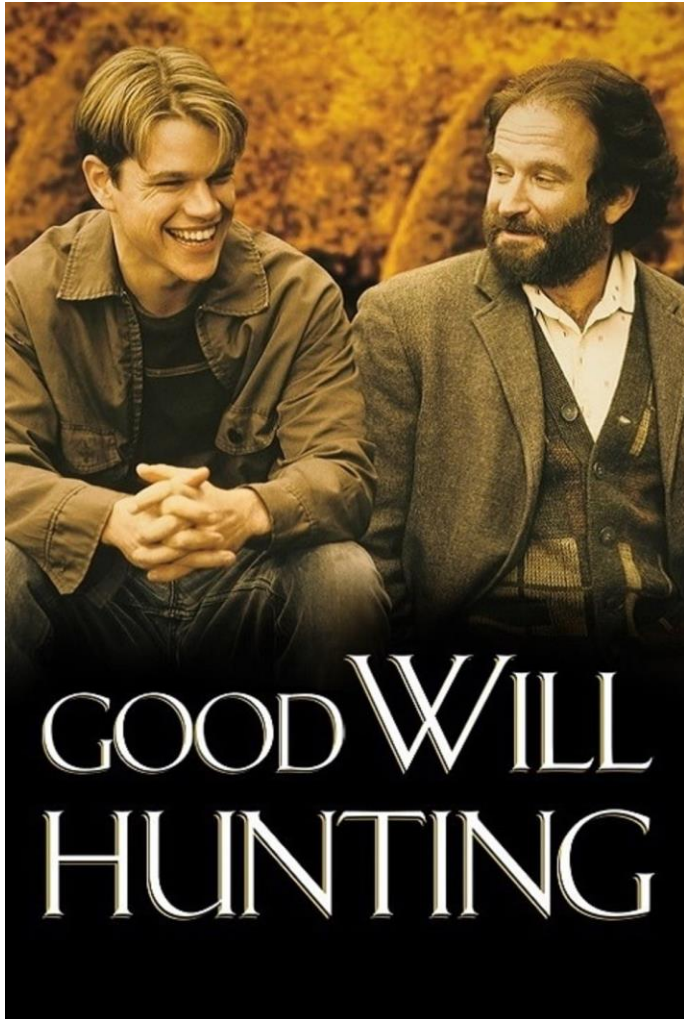


Non-planar circuits



Non-planar circuits

Interesting To watch: Good Will Hunting



- a) An 1997 drama film directed by Gus Van Sant.
- b) 1998 Oscar Best original screenplay and Best supporting Actor
- c) Cast: Matt Damon, Ben Affleck, Casey Affleck, Minnie Driver, Robin Williams, Stellan Skarsgard

Lesson 3 Systematic approaches for circuit analysis

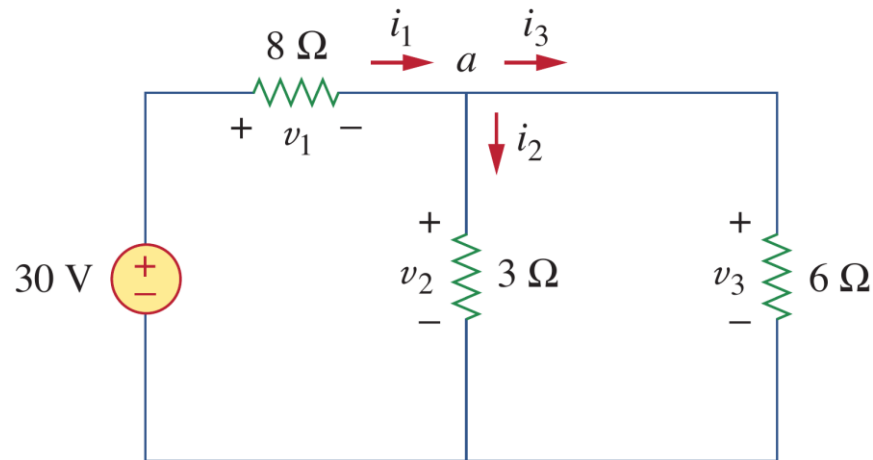
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3.12	Summary and assignments	Slides 84-86

3.3 Introduction (a)

- Have a look at this example we solved in Lesson 2

Find the currents and voltages and in the circuit shown below.

Solution I:



$$\begin{cases} i_1 - i_2 - i_3 = 0 \\ -30 + v_1 + v_2 = 0 \\ -v_2 + v_3 = 0 \\ v_1 = 8i_1 \\ v_2 = 3i_2 \\ v_3 = 6i_3 \end{cases}$$



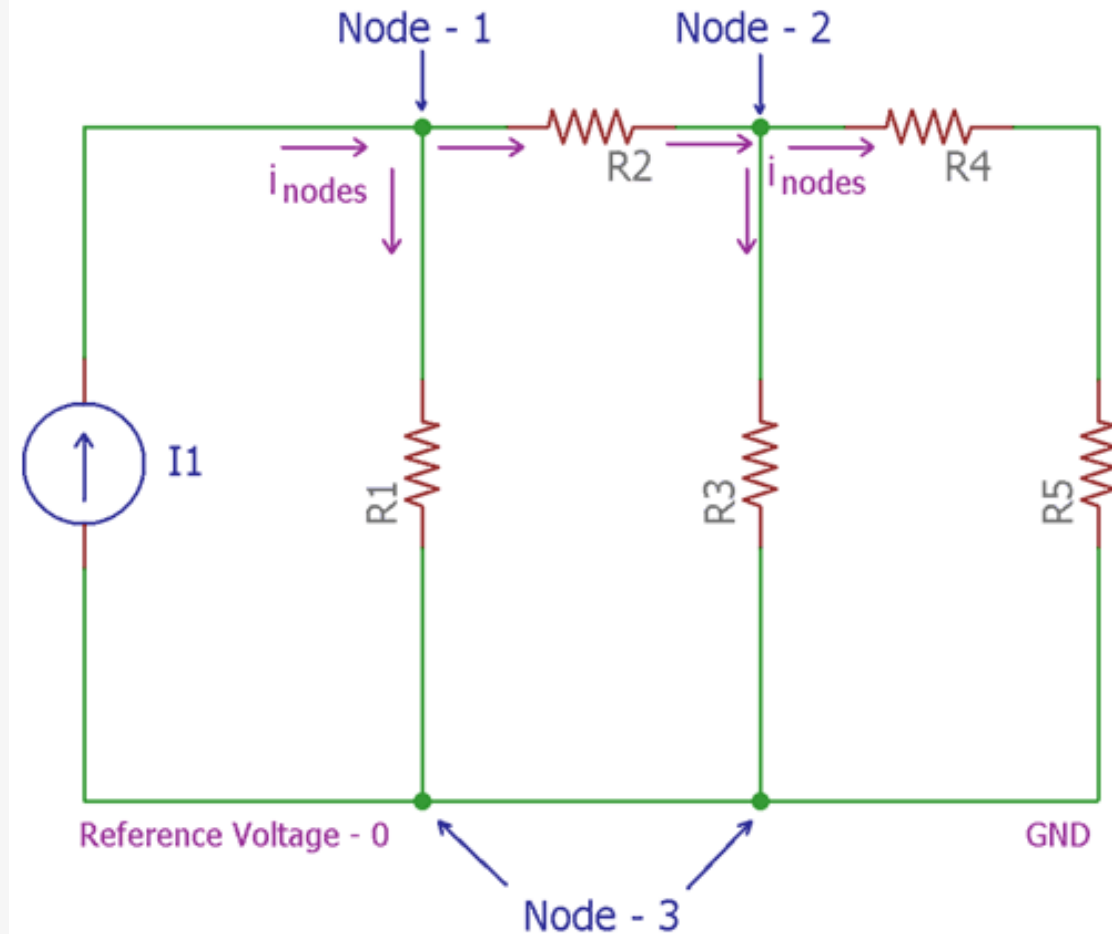
$$\begin{cases} i_1 = 3 \text{ A} \\ i_2 = 2 \text{ A} \\ i_3 = 1 \text{ A} \\ v_1 = 24 \text{ V} \\ v_2 = 6 \text{ V} \\ v_3 = 6 \text{ V} \end{cases}$$

One needs to solve **six-dimensional** linear equations when choosing $v_1, v_2, v_3, i_1, i_2, i_3$ as unknown variables

3.4 Nodal voltage analysis (a)

- Nodal voltage analysis

- Nodal voltage analysis is a circuit analysis technique, which is also called nodal analysis.
- Nodal voltages are selected as the independent variables for the description of the status of an electric circuit, which is why it's called nodal voltage approach.
- Nodal analysis is based on Kirchhoff's Current Law (KCL) with coordination of Ohm's law.
- Nodal analysis can reduce the amount of time required to get important information on a circuit by applying KCL in a step-by-step approach to solve for unknown voltages and current.



3.4 Nodal voltage analysis (b)

- Steps to apply nodal analysis

Step 01

Label the Nodes

- ❑ Select a reference node.
- ❑ Prefer ground as the reference
- ❑ Name voltages at each node.

Step 02

KCL at nodes

- ❑ Apply KCL at each node except the reference node
- ❑ Write down the equation at each node

Step 03

Ohm's Law

- ❑ Apply Ohm's Law to express each current
- ❑ Conventionally current flows from higher to lower potential

Step 04

Solve equations

- ❑ Solve the equations obtained in previous steps.
- ❑ Number of equations will be equal to the number of non-reference nodes.

3.4 Nodal voltage analysis (c)

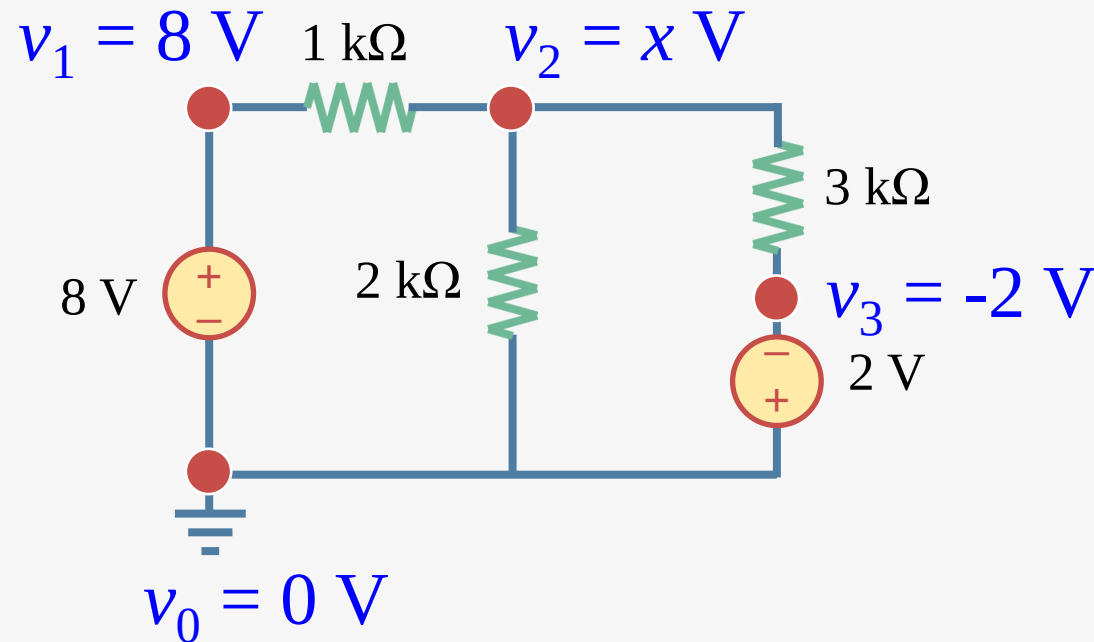
- An example of how to apply nodal analysis

Step 01

Label the Nodes

- Select a reference node.
- Prefer ground as the reference
- Name voltages at each node.

Example 01 Find the currents in the branches



(1) Identify the nodes.

(2) Pick a node as the voltage reference, label it as V_0 . GND is preferred as the reference.

(3) Pick a unlabelled node and label it as v_1 , then pick another unlabeled node and label it as v_2 , until all $(n-1)$ non-reference nodes are labelled.

3.4 Nodal voltage analysis (d)

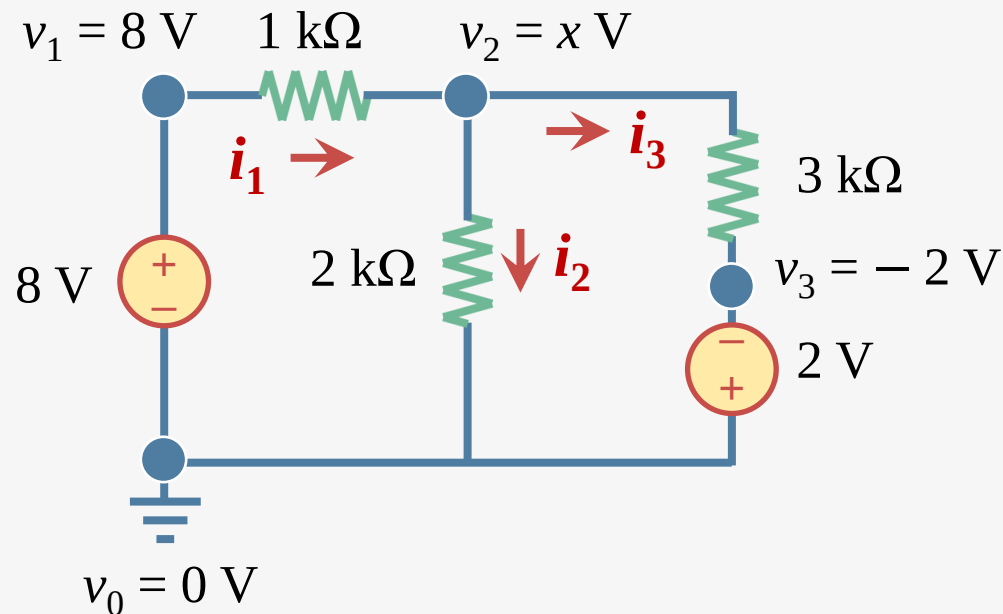
- An example of how to apply nodal analysis

Step 02

KCL at nodes

- Apply KCL at each node except the reference node
- Write out the equation at each node

Example 01 Find the currents in the branches



Apply KCL to node 2

- Select reference current directions if not given.
- Write the KCL equation.

$$i_1 - i_2 - i_3 = 0$$

3.4 Nodal voltage analysis (e)

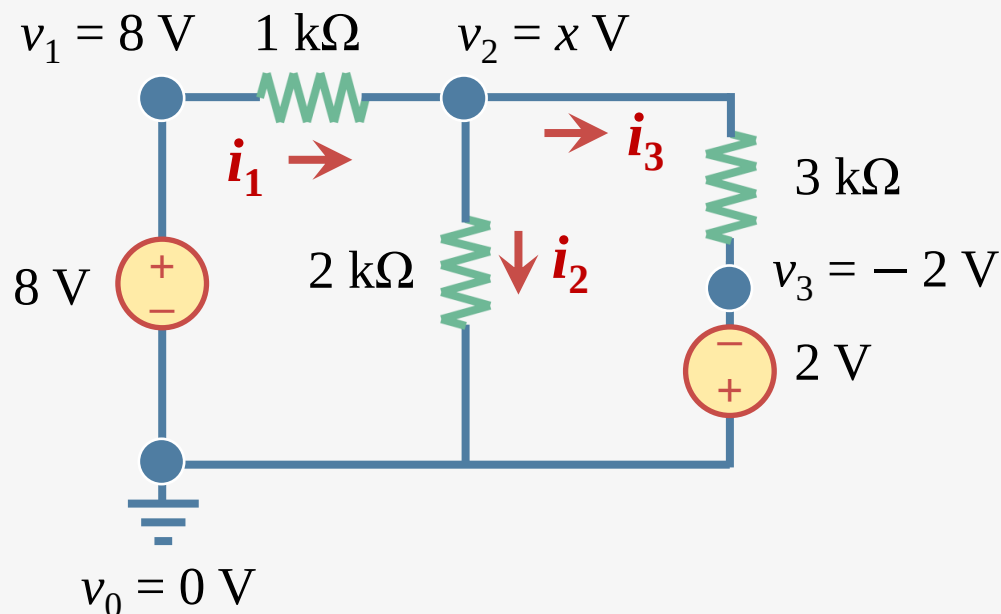
- An example of how to apply nodal analysis

Step 03

Ohm's Law

- Apply Ohm's Law to express each current
- Conventionally current flows from higher to lower potential

Example 01 Find the currents in the branches



Why $(8 - x)$ instead of $(x - 8)$

(6) Use Ohm's law

$$i_1 = (8 - x)/1\text{k}$$

$$i_2 = (x - 0)/2\text{k}$$

$$i_3 = (x - (-2))/3\text{k}$$

(7) Update the KCL equation.

$$i_1 - i_2 - i_3 = 0$$

$$\frac{8 - x}{1\text{k}} - \frac{x - 0}{2\text{k}} - \frac{x - 0}{3\text{k}} = 0$$

3.4 Nodal voltage analysis (g)

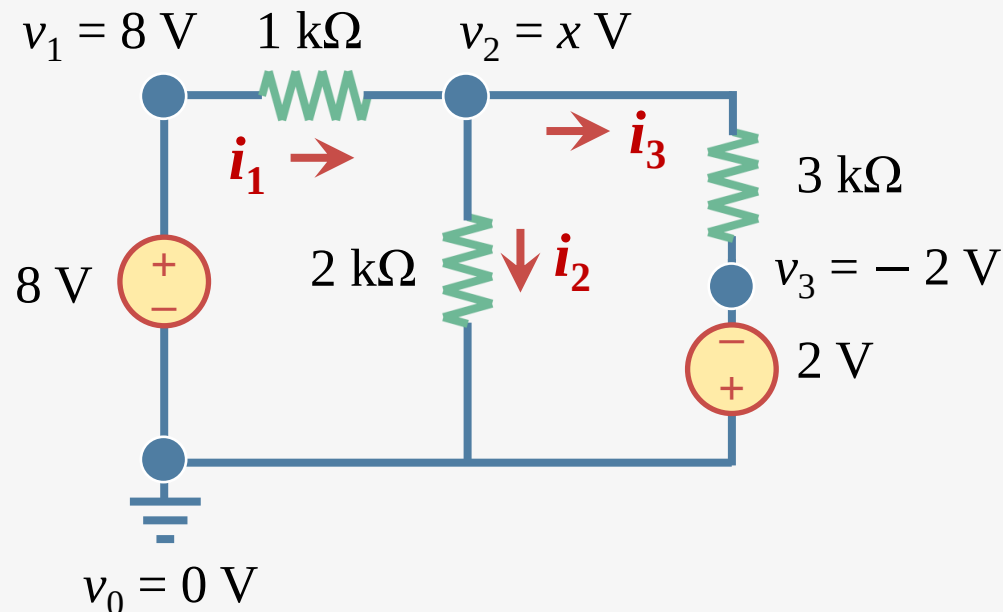
- An example of how to apply nodal analysis

Step 04

Solve equations

- Solve the equations obtained in previous steps.
- Number of equations will be equal to the number of non-reference nodes.

Example 01 Find the currents in the branches



(8) Solve the equation

$$\frac{8 - x}{1k} - \frac{x - 0}{2k} - \frac{x - 0}{3k} = 0$$



$$x = 4 \text{ V}$$



$$i_1 = (8 - x)/1k = 4 \text{ mA}$$

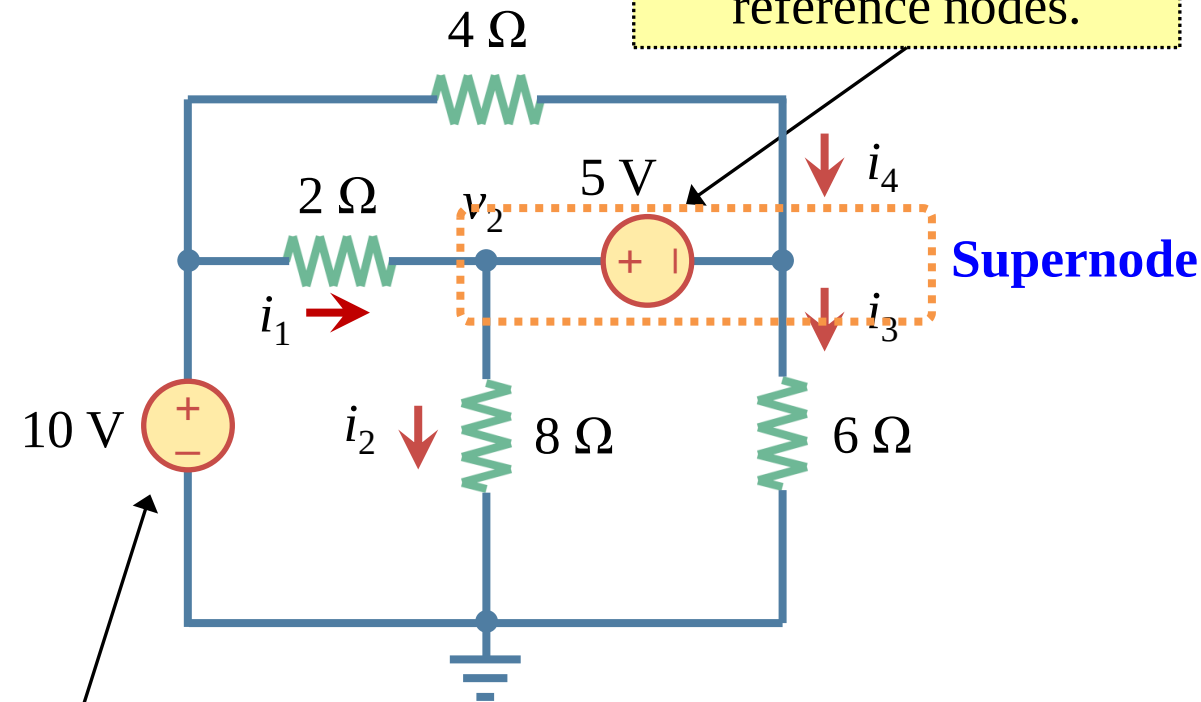
$$i_2 = (x - 0)/2k = 2 \text{ mA}$$

$$i_3 = (x - (-2))/3k = 2 \text{ mA}$$

3.5 Supernodes (a)

- Circuits with floating voltage sources

- Nodal analysis can be difficult when there are voltage sources connected between two non-reference nodes (floating), since the voltage of the voltage source is unknown and cannot be expressed with nodal voltage.
- To address this, the concept of **supernodes** (超节点) are created, which actually reflects the fact that KCL can be applied for a closed boundary in lumped circuits.



Case 1: the V-source connects to the reference node. The voltage at the other terminal of the source is specified. No equation is needed.

3.5 Supernodes (b)

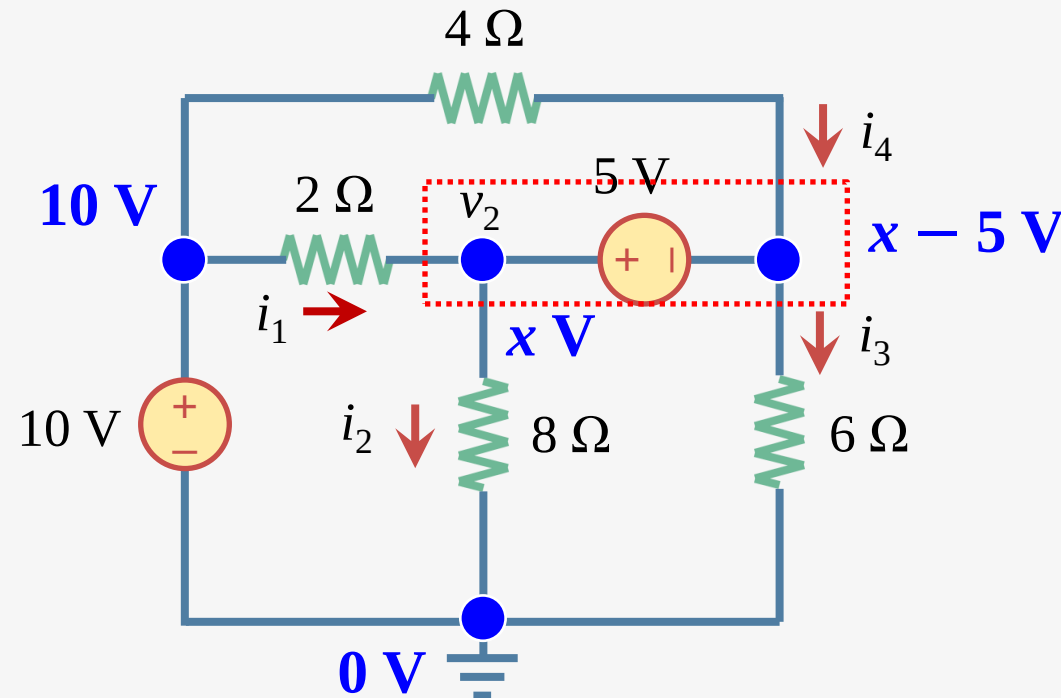
• Supernode

Step 01

Label the Nodes

- ❑ Select a reference node.
- ❑ Prefer ground as the reference
- ❑ Name voltages at each node.

Example 02 Find v_2 in the following circuit



(1) Identify the nodes.

(2) Pick a node as the voltage reference. GND is preferred as the reference.

(3) Notice a supernode is needed for the 5 V voltage source

(4) Pick a unlabelled node and label it as v_1 , then pick another unlabeled node and label it as v_2 , until all $(n-1)$ non-reference nodes are labelled.

3.5 Supernodes (c)

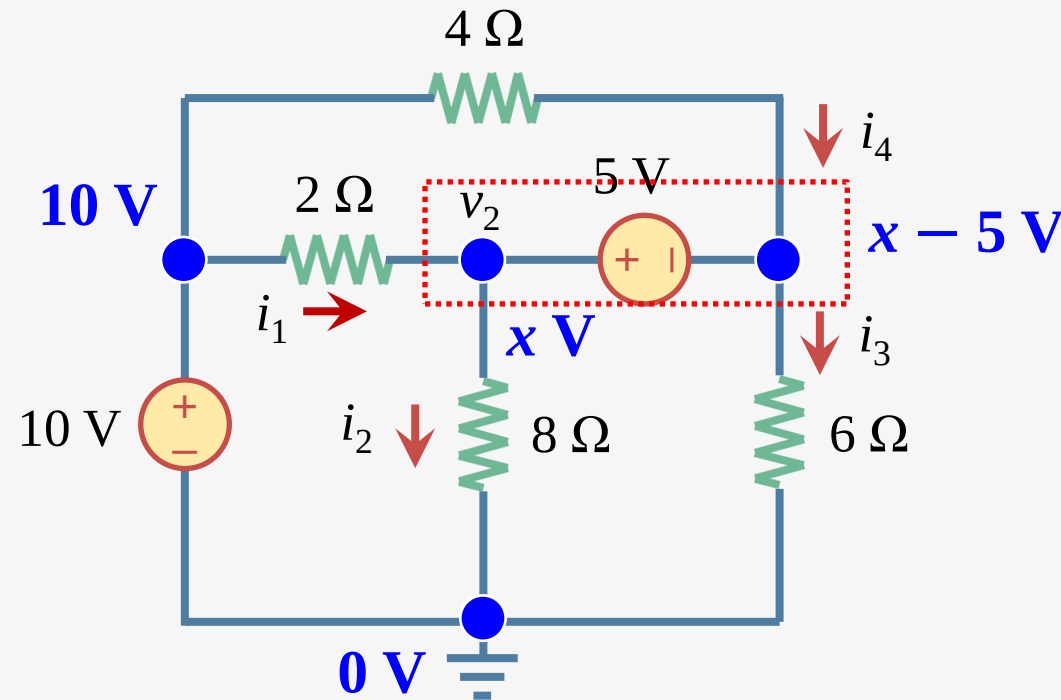
- Supernode

Step 02

KCL at nodes

- Apply KCL at each node except the reference node
- Write out the equation at each node

Example 02 Find v_2 in the following circuit



Apply KCL to the supernode

(5) Here the reference current directions are given, so you do not need to select new ones.

(6) Write the KCL equation.

$$i_1 - i_2 - i_3 + i_4 = 0$$

3.5 Supernodes (d)

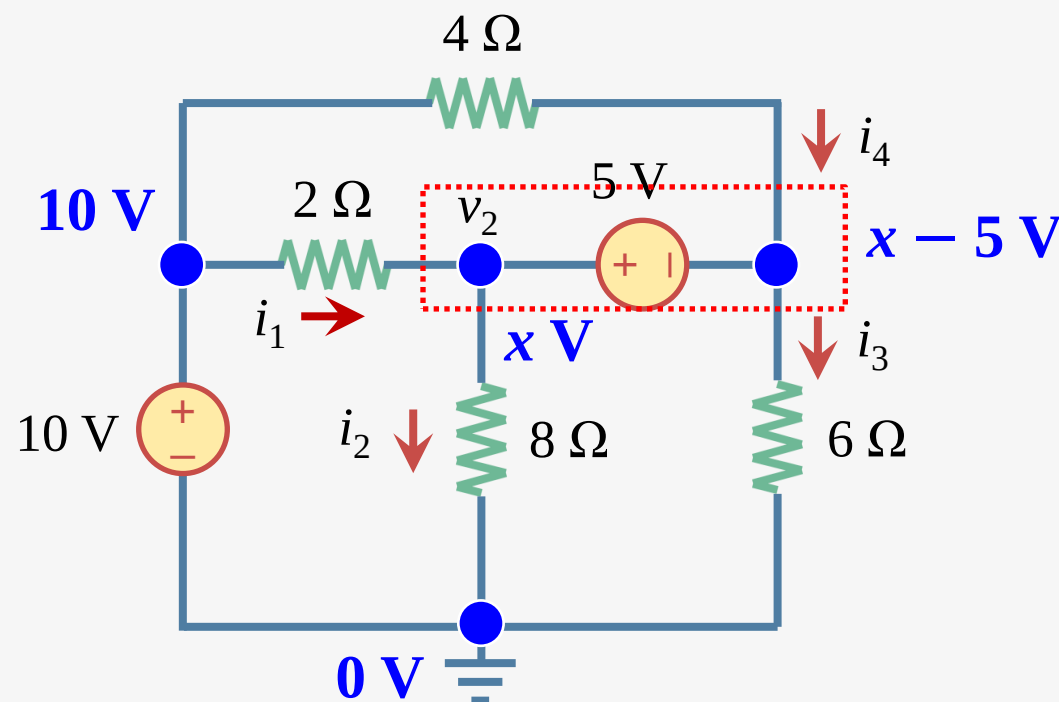
• Supernode

Step 03

Ohm's Law

- Apply Ohm's Law to express each current
- Conventionally current flows from higher to lower potential

Example 02 Find v_2 in the following circuit



(7) Use Ohm's law

$$i_1 = (10 - x)/2$$

$$i_2 = (x - 0)/8$$

$$i_3 = ((x - 5) - 0)/6$$

$$i_4 = (10 - (x - 5))/4$$

(8) Update the KCL equation.

$$i_1 - i_2 - i_3 + i_4 = 0$$

$$\frac{10 - x}{2} - \frac{x}{8} - \frac{x - 5}{6} + \frac{15 - x}{4} = 0$$

3.5 Supernodes (e)

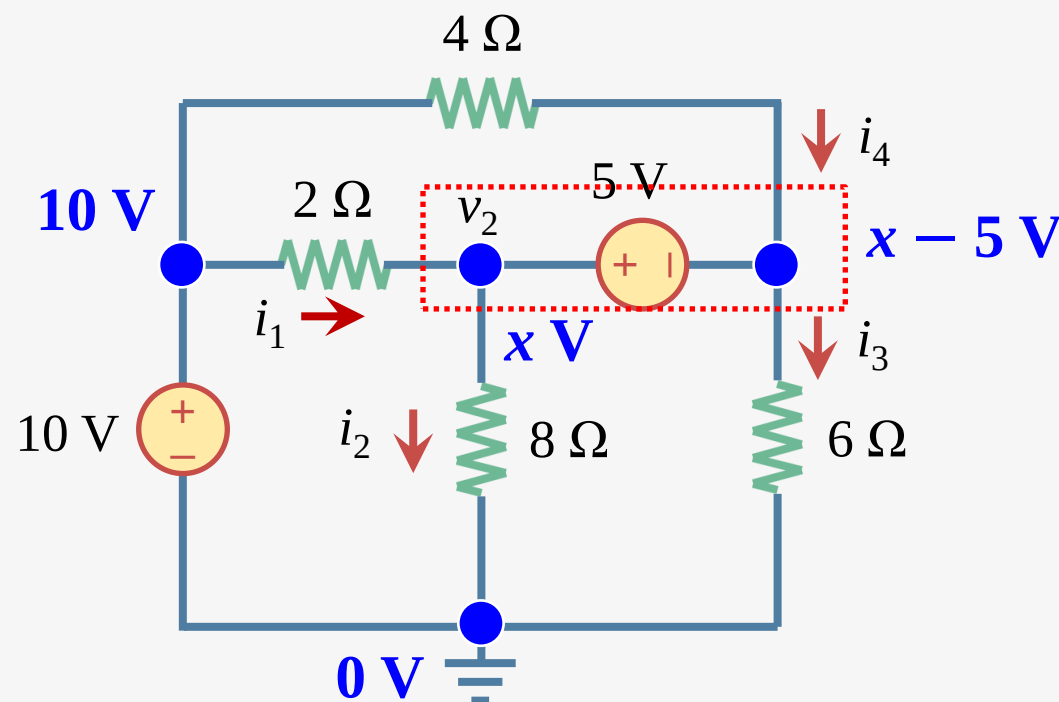
• Supernode

Step 04

Solve equations

- Solve the equations obtained in previous steps.
- Number of equations will be equal to the number of non-reference nodes.

Example 02 Find v_2 in the following circuit



(9) Solve the equation.

$$\frac{10 - x}{2} - \frac{x}{8} - \frac{x - 5}{6} + \frac{15 - x}{4} = 0$$

$$v_2 = x = 9.2 \text{ V}$$

3.6 More examples (a)

- More examples for nodal analysis

Step 01
Label the Nodes
Step 02
KCL at nodes
Step 03
Ohm's Law
Step 04
Solve equations

In the following slides we will see how to apply nodal analysis to more types of circuits, including:

- a) Circuits with current source
- b) Weighted average circuits
- c) Digital-to-analog converters
- d) Circuits with dependent current sources
- e) Circuits with dependent voltage sources

3.6 More examples (b)

- Circuits with current sources

Example 03: Find the currents and voltages in the circuit shown below.

Step 01

Label the Nodes

Step 02

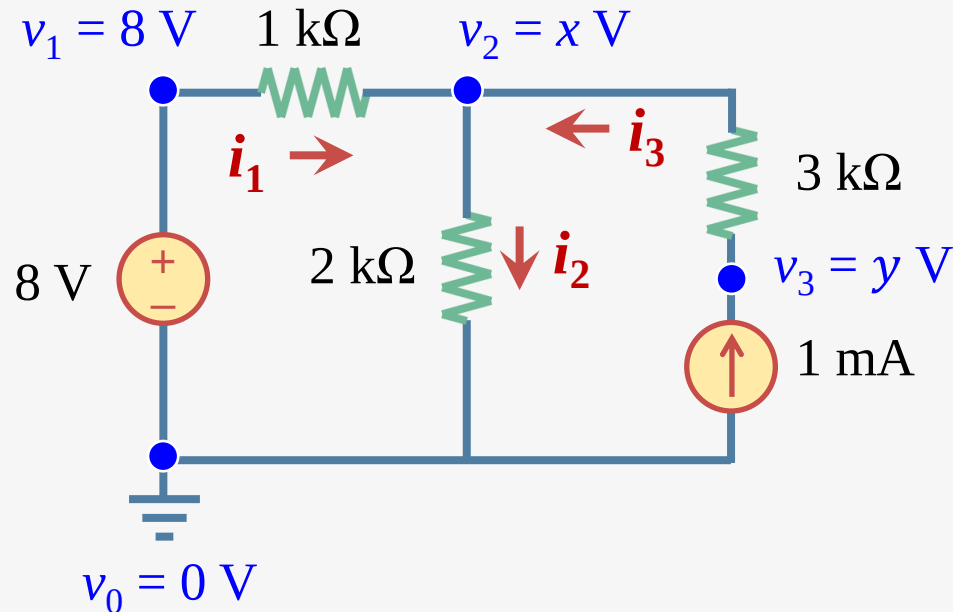
KCL at nodes

Step 03

Ohm's Law

Step 04

Solve equations



Solution:

$$\begin{cases} \frac{8 - x}{1k} - \frac{x - 0}{2k} + \frac{y - x}{3k} = 0 \\ -\frac{y - x}{3k} + 1 \text{ mA} = 0 \end{cases}$$

$$v_2 = x = 6 \text{ V} \quad \& \quad v_3 = y = 9 \text{ V}$$

$$i_1 = (8 - 6)/1k = 2 \text{ mA}$$

$$i_2 = (x - 0)/2k = 3 \text{ mA}$$

3.6 More examples (c)

- Weighted average circuits

Example 04: Find the v_{out} in the circuit shown below. $v_1, v_2, v_3, R_1, R_2, R_3$ are known.

Step 01

Label the Nodes

Step 02

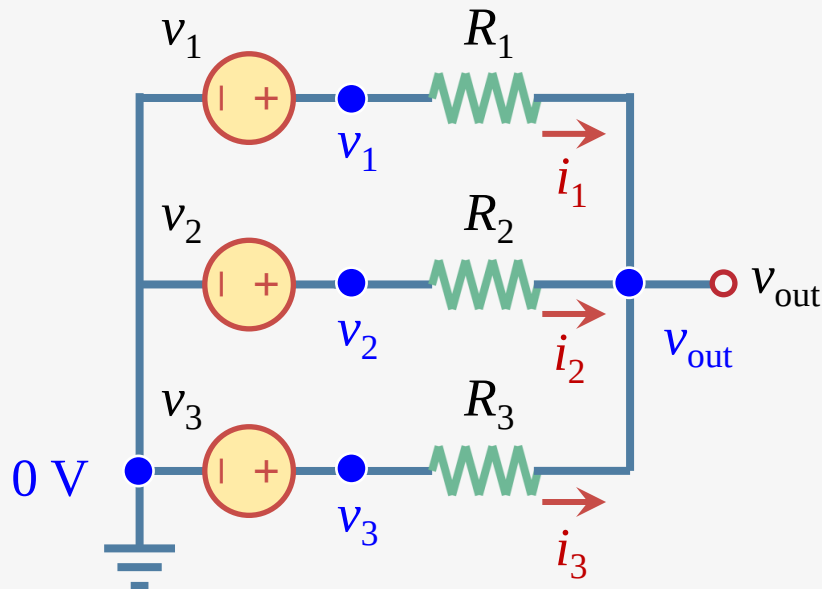
KCL at nodes

Step 03

Ohm's Law

Step 04

Solve equations



*A very useful sub-circuit that calculates the weighted average of any number of voltages.

Solution:

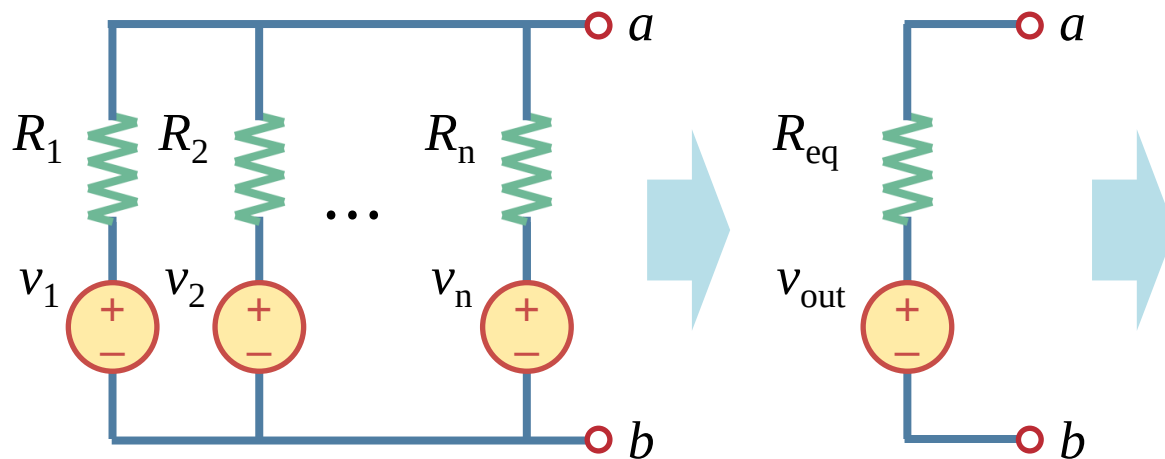
$$\frac{v_1 - v_{\text{out}}}{R_1} + \frac{v_2 - v_{\text{out}}}{R_2} + \frac{v_3 - v_{\text{out}}}{R_3} = 0$$

$$(v_1 - v_{\text{out}}) \cdot G_1 + (v_2 - v_{\text{out}}) \cdot G_2 + (v_3 - v_{\text{out}}) \cdot G_3 = 0$$

$$V_{\text{out}} = \frac{G_1 v_1 + G_2 v_2 + G_3 v_3}{G_1 + G_2 + G_3}$$

$$V_{\text{out}} = \frac{v_1 + v_2 + v_3}{3}, \text{ when } R_1 = R_2 = R_3$$

Interesting To Know: Millman's Theorem Equation (弥尔曼定理)



$$v_{out} = \frac{\sum_1^n v_i G_i}{\sum_1^n G_i} = \frac{\frac{v_1}{R_1} + \frac{v_2}{R_2} + \dots + \frac{v_n}{R_n}}{\frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}}$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$$

- In electrical engineering, Millman's theorem (or the parallel generator theorem) is a method to simplify the solution of a circuit. Specifically, Millman's theorem is used to compute the voltage at the ends of a circuit made up of only branches in parallel
- The Millman's Theorem states that – when a number of voltage sources are in parallel **having internal resistance (or being in series with resistors)** respectively, the arrangement can replace by a single equivalent voltage source v_{out} in series with an equivalent series resistance R_{eq} .

3.6 More examples (d)

- Digital to analog converters (DAC)

Example 05: Find the v_{out} in the following circuit.

Step 01	
Label the Nodes	
Step 02	
KCL at nodes	
Step 03	
Ohm's Law	
Step 04	
Solve equations	

*Weighted resistor DACs

More information:

- A 3-bit binary number, $b = b_2b_1b_0$, has bit-weights of 4, 2 and 1.
- If we label the bits $b_2b_1b_0$, then $b = 4b_2 + 2b_1 + b_0$ in decimal
- Thus 110 has a value of 6 in decimal
- $b_2b_1b_0$ are used to control the switches which determine whether $v_i = V_{\text{DD}}$ or 0 V. Thus $v_i = V_{\text{DD}} \cdot b_i$.

3.6 More examples (e)

- Digital to analog converters

Example 05: Find the v_{out} in the following circuit.

Step 01 Label the Nodes	
Step 02 KCL at nodes	
Step 03 Ohm's Law	
Step 04 Solve equations	

*Weighted resistor DACs

Solutions:

From example 04 we have:

$$V_{\text{out}} = \frac{G_1 v_1 + G_2 v_2 + G_3 v_3}{G_1 + G_2 + G_3}$$

$$V_{\text{out}} = \frac{\frac{1}{2} v_1 + \frac{1}{4} v_2 + \frac{1}{8} v_3}{\frac{1}{2} + \frac{1}{4} + \frac{1}{8}}$$

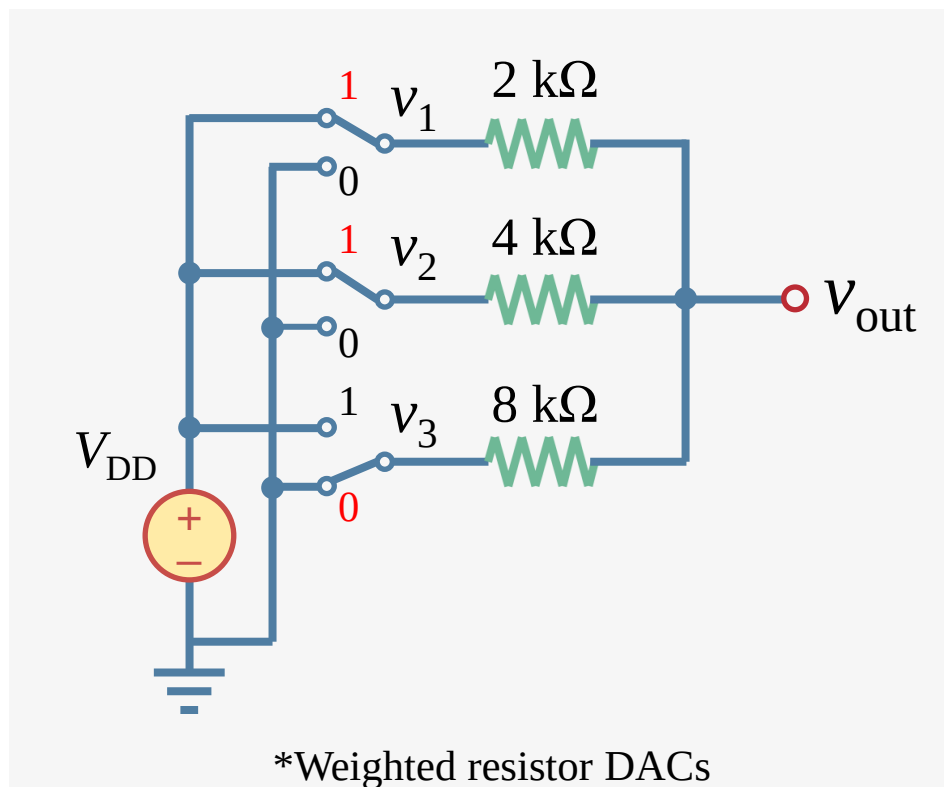
$$V_{\text{out}} = \frac{(4b_2 + 2b_1 + b_0)}{7} \cdot V_{\text{DD}} = \frac{b}{7} \cdot V_{\text{DD}}$$

3.6 More examples (f)

- Digital to analog converters

Example 05: Find the v_{out} in the following circuit.

Step 01
Label the Nodes
Step 02
KCL at nodes
Step 03
Ohm's Law
Step 04
Solve equations



Solutions:

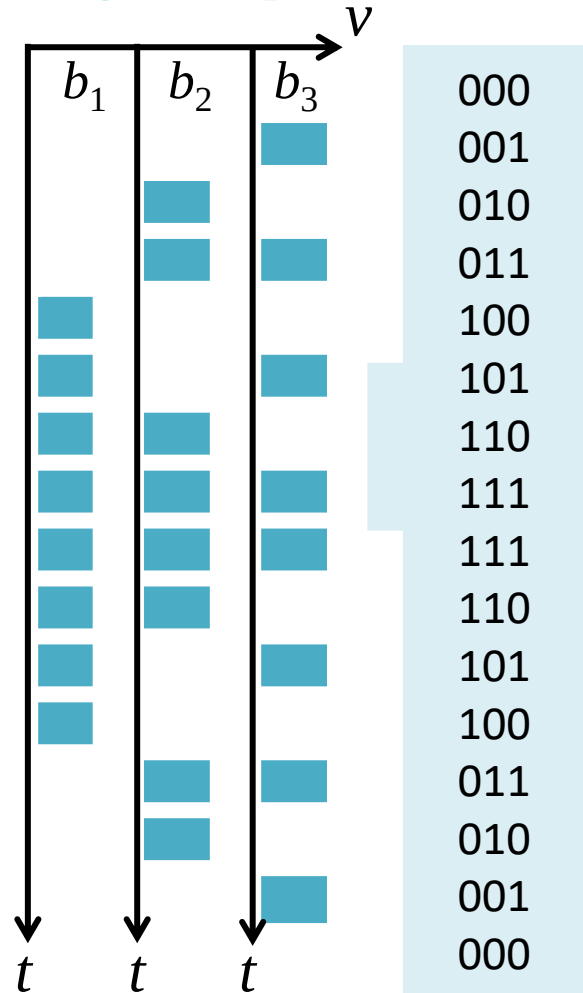
From example 02 we have:

$$V_{\text{out}} = \frac{(4b_2 + 2b_1 + b_0)}{7} \cdot V_{\text{DD}} = \frac{b}{7} \cdot V_{\text{DD}}$$

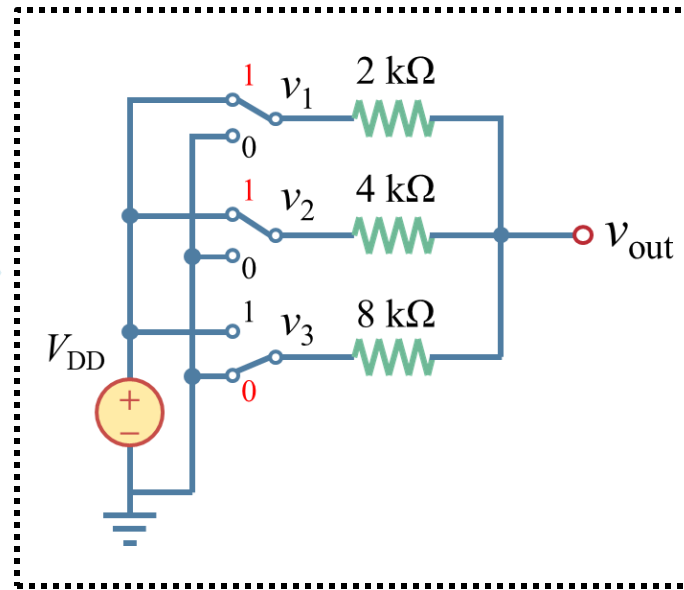
Input (D)		Output (A)
$b_2b_1b_0$ (binary)	b (decimal)	v_{out}
000	0	0
001	1	$1/7 \cdot V_{\text{DD}}$
010	2	$2/7 \cdot V_{\text{DD}}$
011	3	$3/7 \cdot V_{\text{DD}}$
100	4	$4/7 \cdot V_{\text{DD}}$
101	5	$5/7 \cdot V_{\text{DD}}$
110	6	$6/7 \cdot V_{\text{DD}}$
111	7	V_{DD}

3.6 More examples (g)

Digital input



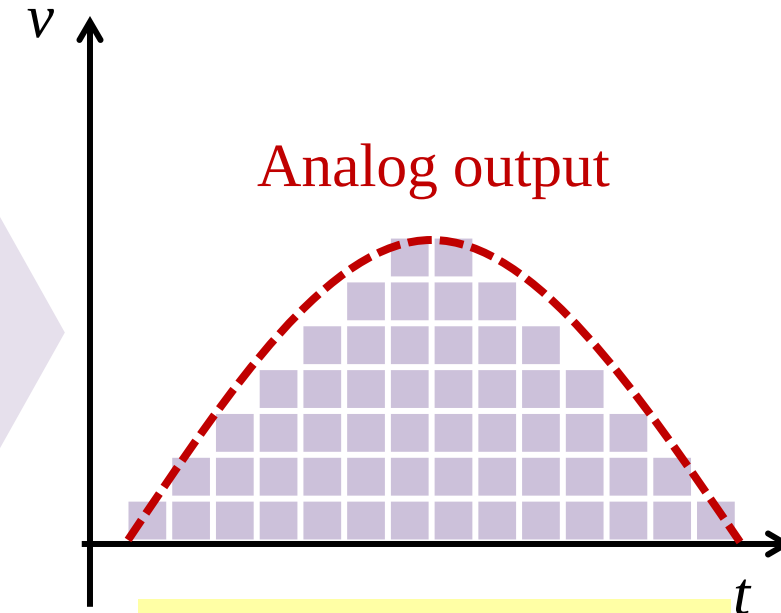
A DAC you've just designed



To improve accuracy of conversion:

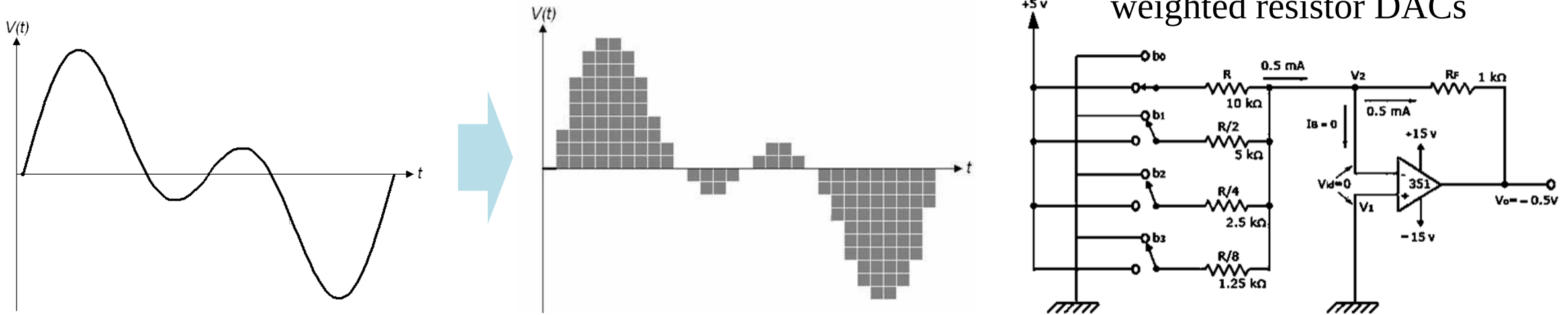
- Increase the number of bits
- Higher sampling rates

$0/7 \cdot V_{DD}$
 $1/7 \cdot V_{DD}$
 $2/7 \cdot V_{DD}$
 $3/7 \cdot V_{DD}$
 $4/7 \cdot V_{DD}$
 $5/7 \cdot V_{DD}$
 $6/7 \cdot V_{DD}$
 $7/7 \cdot V_{DD}$
 $7/7 \cdot V_{DD}$
 $6/7 \cdot V_{DD}$
 $5/7 \cdot V_{DD}$
 $4/7 \cdot V_{DD}$
 $3/7 \cdot V_{DD}$
 $2/7 \cdot V_{DD}$
 $1/7 \cdot V_{DD}$
 $0/7 \cdot V_{DD}$



Interesting to think: what are the disadvantages of weighted resistor DACs?

Interesting To Know: digital to analog converters (DACs)



- a) In electronics, a digital-to-analog converter (DAC) is a system that converts a digital signal into an analog signal. An analog-to-digital converter performs the reverse function.
- b) The digital signal such as the binary signal exist in the form of bits, and it is the combination of 1's & 0's (or High & low voltage levels). The DAC converts these bits into an analog voltage or current.
- c) There are two types of DACs, weighted resistor DACs and R-2R ladder DACs.

Interesting To think: CD vs Vinyl Record

CD (digital)



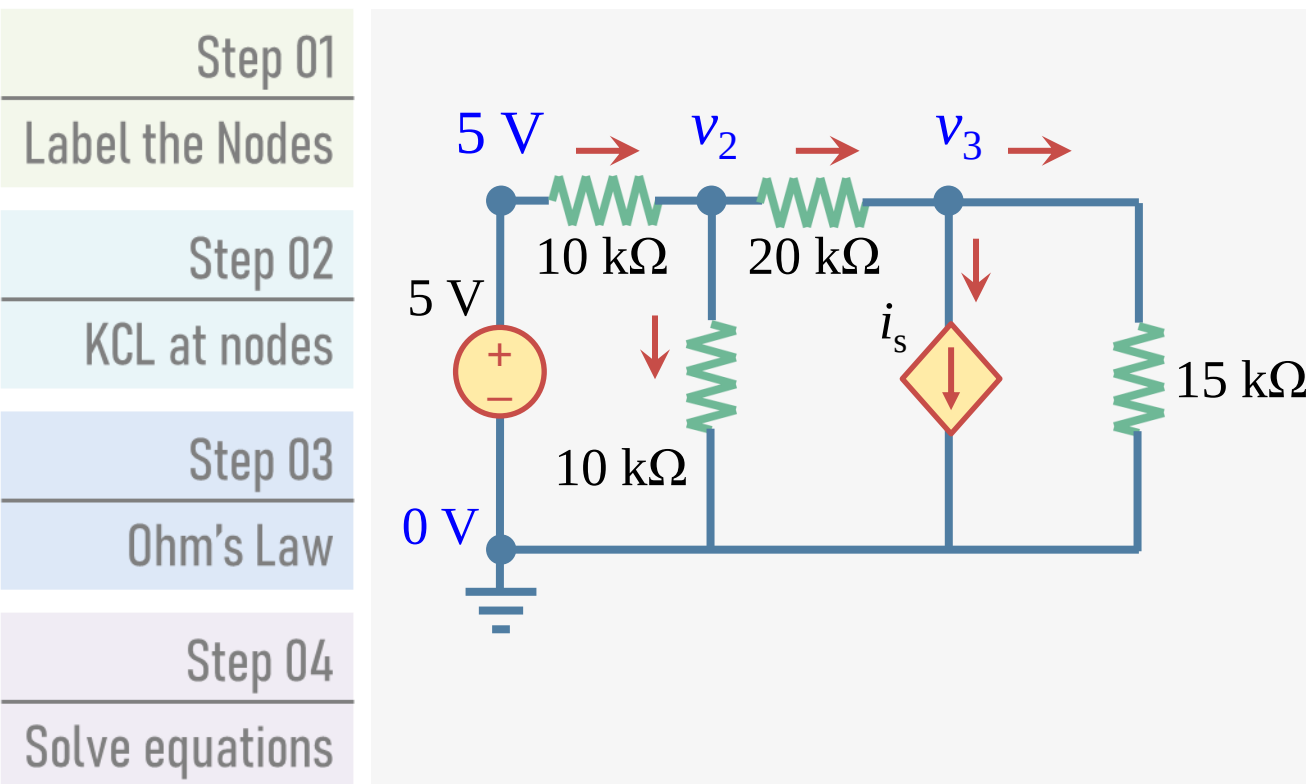
Vinyl Record



3.6 More examples (h)

- Circuits with **dependent current sources**

Example 06: Find the v_2 , v_3 , i_s in the following circuit. $i_s = 0.2(5 - v_2)$.



Solution:

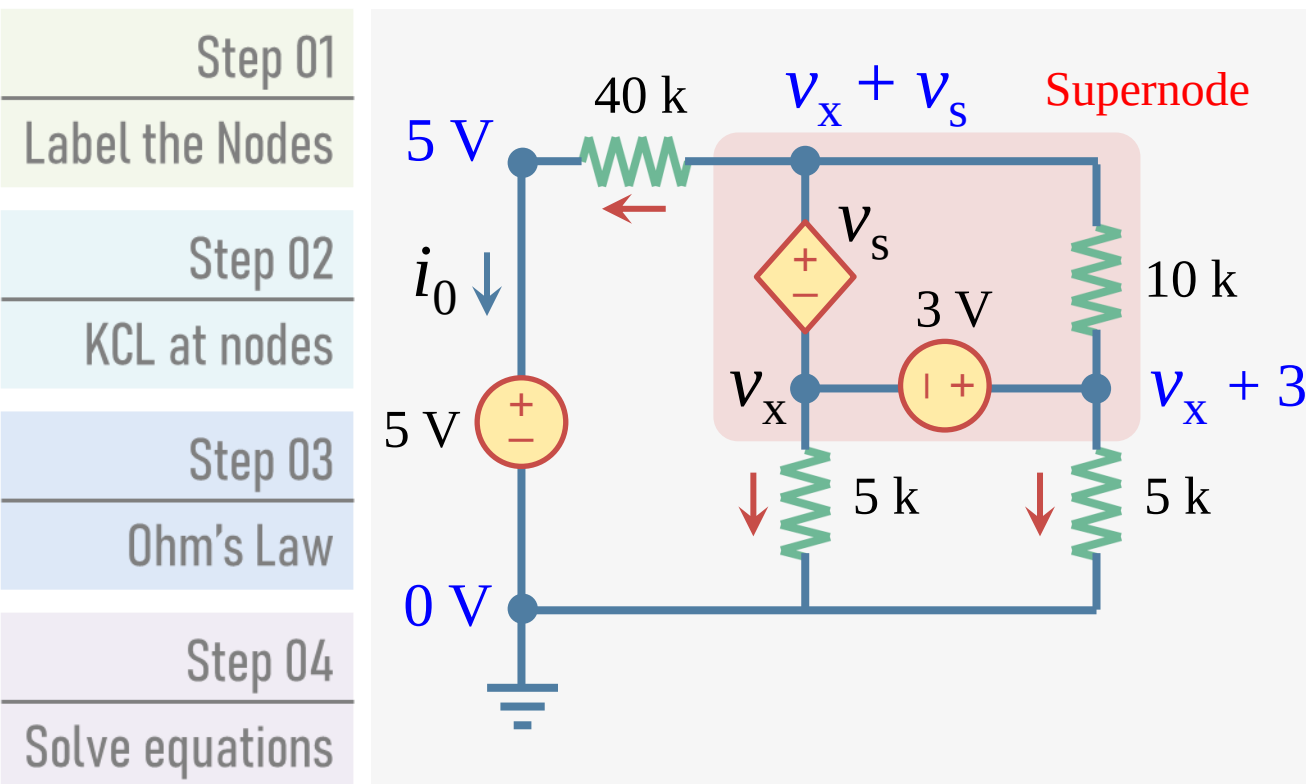
$$\begin{cases} \frac{5 - v_2}{10k} - \frac{v_2 - 0}{10k} - \frac{v_2 - v_3}{20k} = 0 \\ \frac{v_2 - v_3}{20k} - \frac{v_3 - 0}{15k} - 0.2(5 - v_2) = 0 \end{cases}$$

$$\begin{cases} v_2 = 0.5 \text{ V} \\ v_3 = -7.5 \text{ V} \\ i_s = 0.9 \text{ mA} \end{cases}$$

3.6 More examples (i)

- Circuits with **dependent voltage sources**

Example 07: Find the v_x and v_s in the following circuit. $v_s = 10 i_0$ and i_0 is in mA.



Solution:

$$\begin{cases} -\frac{v_x + v_s - 5}{40k} - \frac{v_x - 0}{5k} - \frac{v_x + 3 - 0}{5k} = 0 \\ -\frac{v_s}{10k} + \frac{v_x + v_s - 5}{40k} = 0 \end{cases}$$

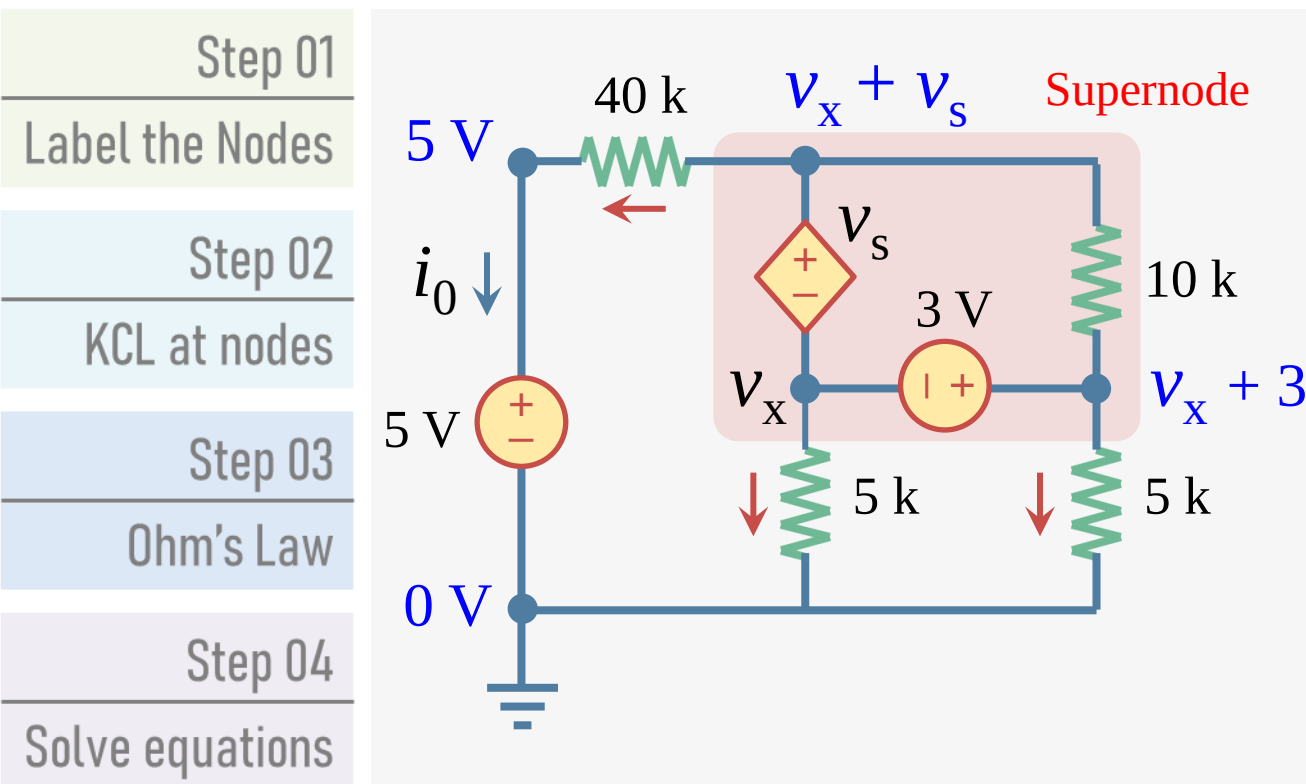
$$\begin{cases} -17v_x - v_s = 19 \\ v_x - 3v_s = 5 \end{cases} \Rightarrow \begin{cases} v_x = -1 \text{ V} \\ v_s = -2 \text{ V} \end{cases}$$

A practice: Could you calculate i_0 ?

3.6 More examples (j)

- Circuits with dependent voltage sources

Example 07: Find the v_x and v_s in the following circuit. $v_s = 10 i_0$ and i_0 is in mA.



Solution:

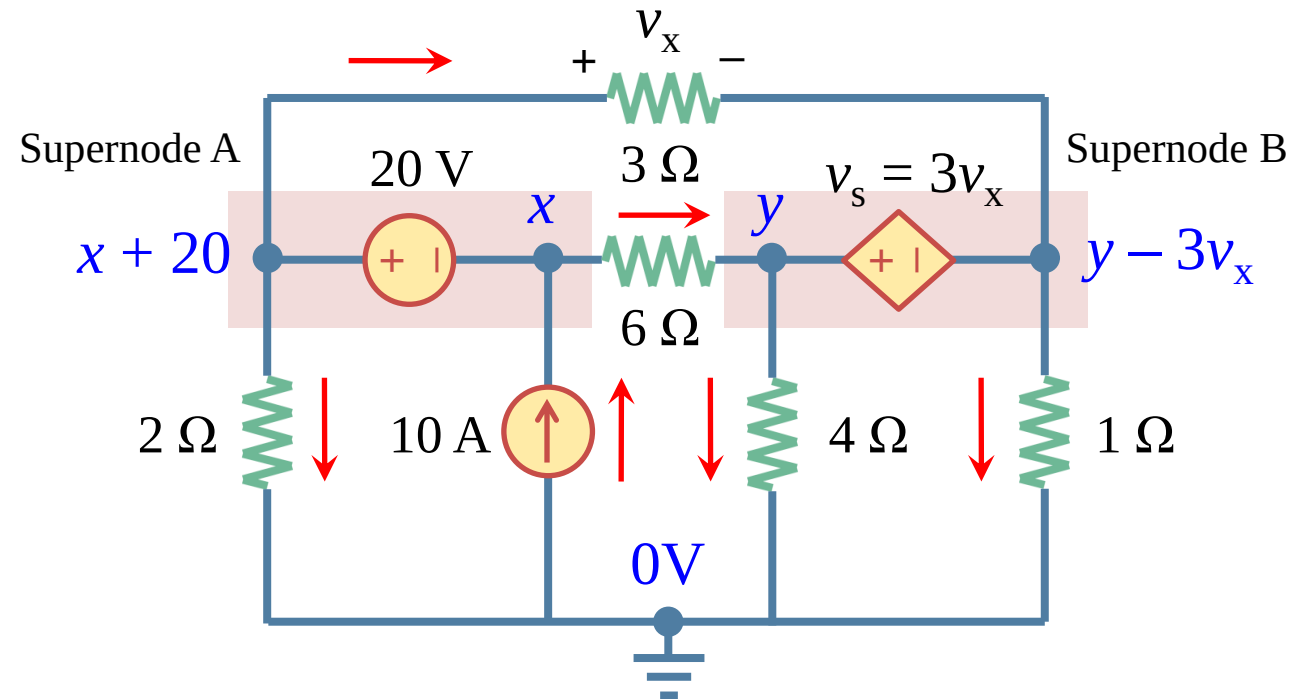
$$\left\{ \begin{aligned} -\frac{v_x + v_s - 5}{40k} - \frac{v_x - 0}{5k} - \frac{v_x + 3 - 0}{5k} &= 0 \\ v_s &= 10i_0 = 10 \times \frac{v_s + v_s - 5}{40k} \end{aligned} \right.$$

$$\left\{ \begin{aligned} -17v_x - v_s &= 19 \\ v_x - 3v_s &= 5 \end{aligned} \right. \Rightarrow \left\{ \begin{aligned} v_x &= -1 \text{ V} \\ v_s &= -2 \text{ V} \end{aligned} \right.$$

You can also get this from the expression of the source values in terms of node voltages

3.6 More examples (h)

Example 08: Find the node voltages in the following circuits.



Solution:

(a) Label the nodes.

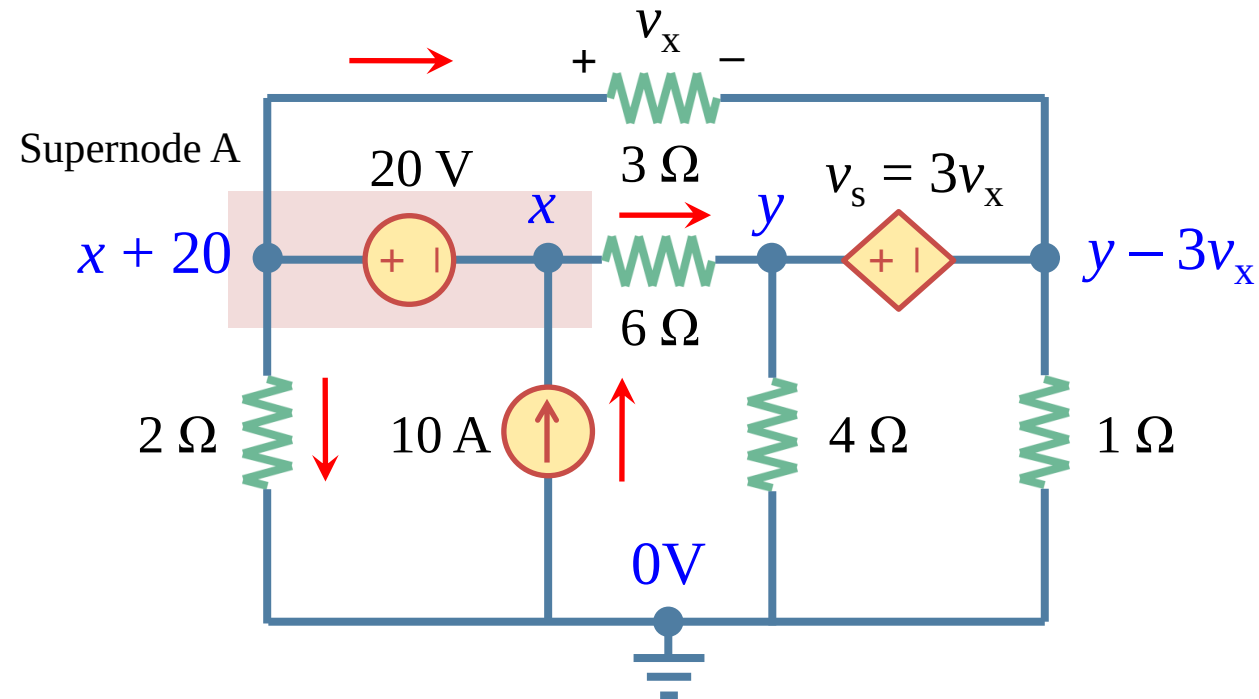
(b) Label the supernodes

(c) KCL equations

- Reference current directions

3.6 More examples (i)

Example 08: Find the node voltages in the following circuits.



Solution:

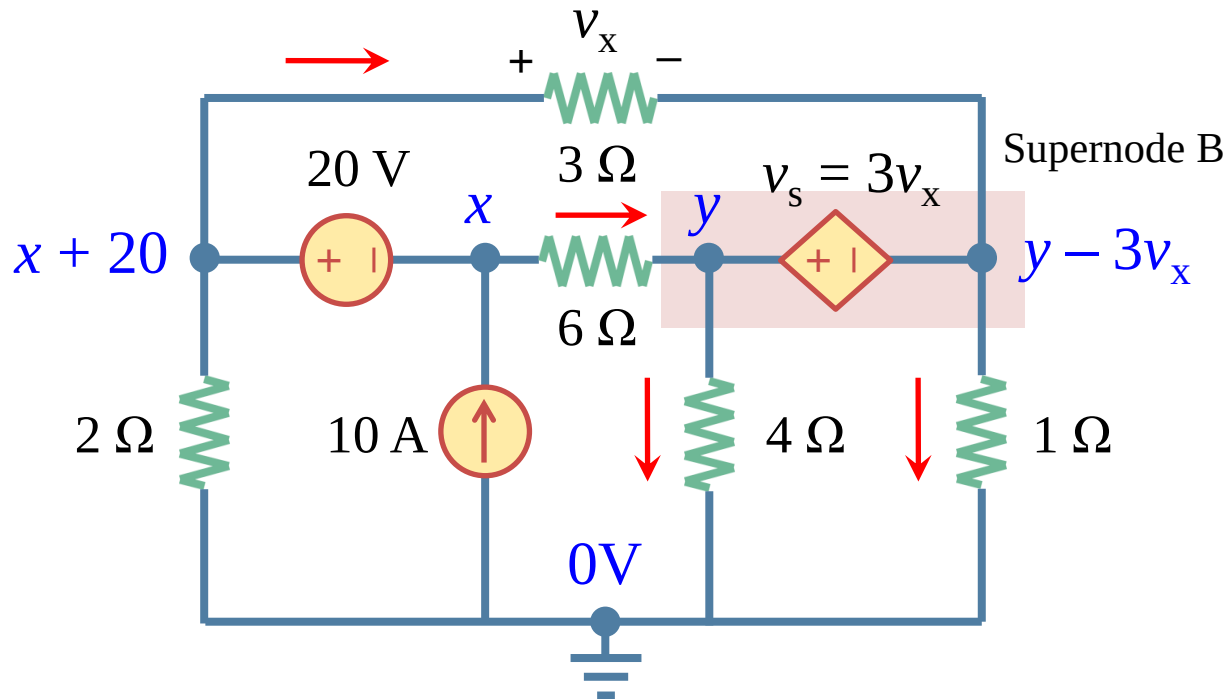
(d) Write an KCL equation for one node

Eq. 1 from Supernode A:

$$-\frac{x + 20}{2} + 10 - \frac{x - y}{6} - \frac{v_x}{3} = 0$$

3.6 More examples (j)

Example 08: Find the node voltages in the following circuits.



Solution:

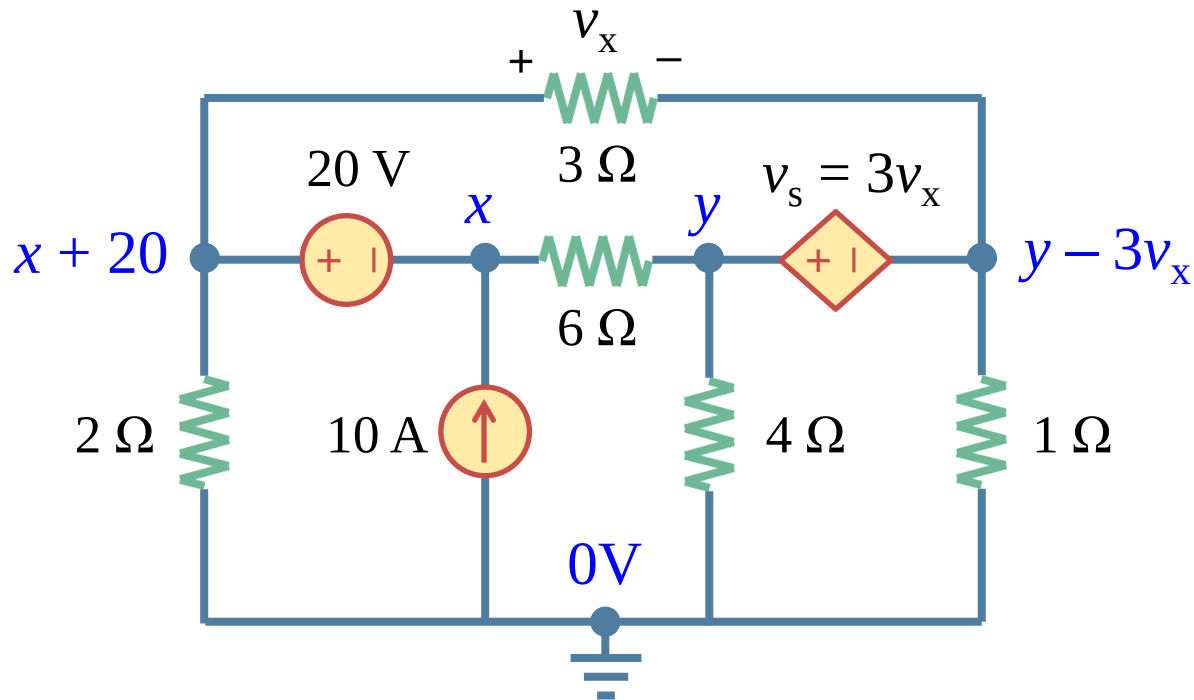
(e) Write an KCL equation for the other node

Eq. 2 from Supernode B:

$$\frac{x - y}{6} - \frac{y - 0}{4} - \frac{y - 3v_x}{1} + \frac{v_x}{3} = 0$$

3.6 More examples (k)

Example 08: Find the node voltages in the following circuits.



Solution:

(f) Rewrite the expression of the dependent sources using nodal voltages.

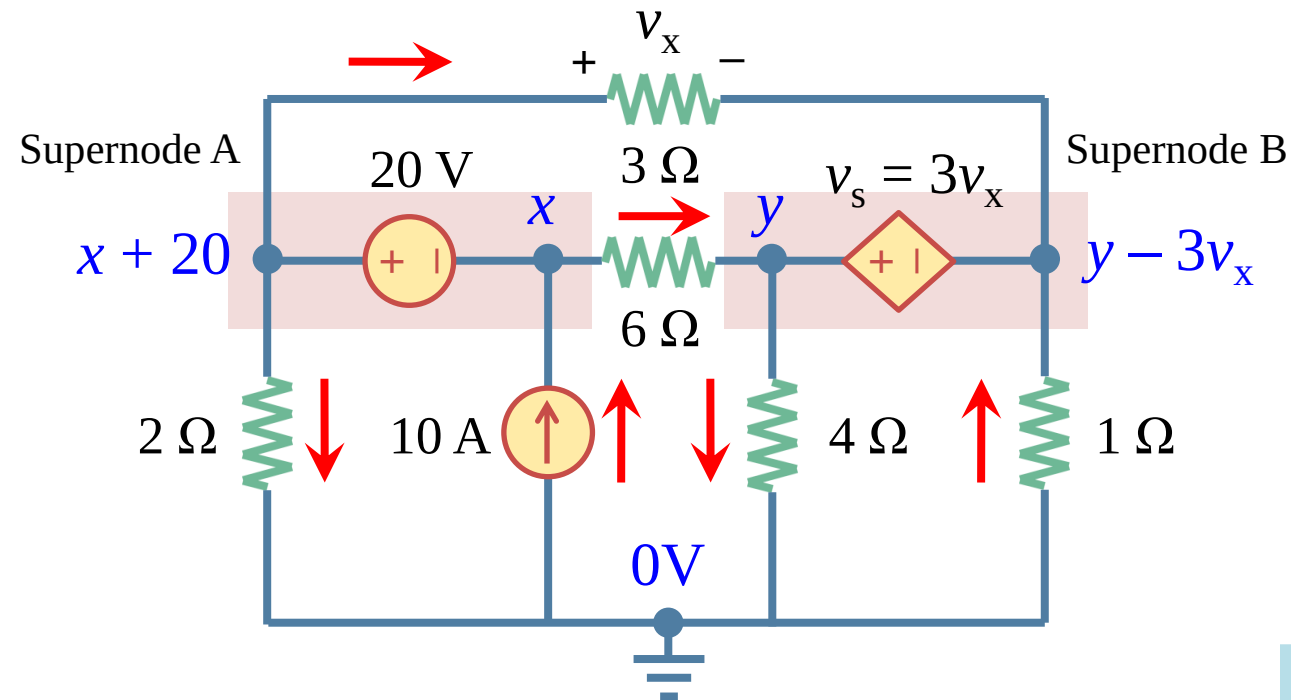
Eq. 3 form the dependent source

$$v_s = 3v_x = 3 \times [x + 20 - (y - 3v_x)]$$

$$x - y + 2v_s = -20$$

3.6 More examples (k)

Example 08: Find the node voltages in the following circuits.



Solution:

Supernode A

$$\begin{cases} -\frac{x+20}{2} + 10 - \frac{x-y}{6} - \frac{v_x}{3} = 0 \\ \frac{x-y}{6} - \frac{y-0}{4} - \frac{y-3v_x}{1} + \frac{v_x}{3} = 0 \\ x - y + 2v_s = -20 \end{cases}$$

Supernode B

The dependent source

$$\begin{aligned} x &= 6.67 \text{ V}, y = 173.33 \text{ V}, v_x = 73.3 \text{ V} \\ x + 20 &= 26.67 \text{ V}, y - 3v_x = -46.67 \text{ V} \end{aligned}$$

3.7 Summary of nodal analysis (a)

- Nodal analysis for different circuits:
 - a) Simple circuits (no floating or dependent voltage sources)
 - b) Floating voltage sources
 - Use supernodes: all the nodes connected by floating voltage sources (independent or dependent)
 - c) Dependent voltage and current sources
 - Label each source with a variable
 - Write extra equations expressing the source values in terms of node voltages
 - Write down the KCL equations as before

3.7 Summary of nodal analysis (b)

- **Universal Nodal Analysis Algorithm:**

Step 01

Label the Nodes

- a) Pick any node as the voltage reference. Label its voltage as 0 V. Label dependent sources with v_s or i_s .
- b) If any voltage sources are connected to a labelled node, label their other ends by adding the value of the source onto the voltage of the labelled end. Repeat as many times as possible.
- c) Pick an unlabelled node and label it, then loop back to step b) until all nodes are labelled.
- d) For each dependent source, write down an equation that expresses its value in terms of other node voltages.

Step 02

KCL at nodes

- e) Write down a KCL equation for each “normal” node (i.e. one that is not connected to a floating voltage source).
- f) Write down a KCL equation for each “super-node”. A super-node consists of a set of nodes that are joined by floating voltage sources and includes any other components joining these nodes.

Step 03

Ohm's Law

- g) Substitute the currents using Ohm's Law

Step 04

Solve equations

- h) Solve the equations

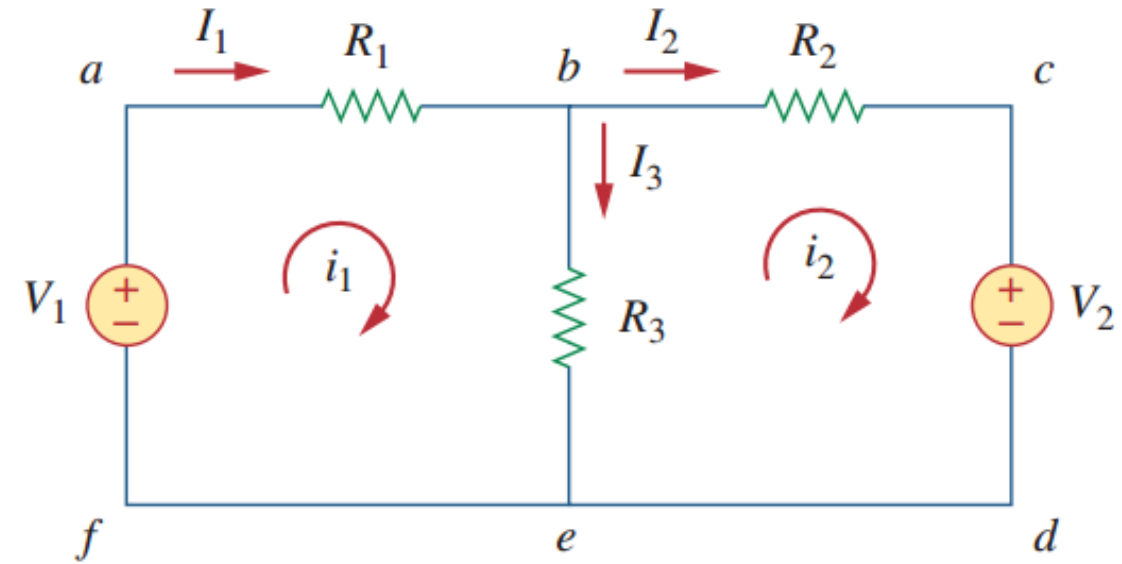
Lesson 3 Systematic approaches for circuit analysis

3.1-3.2	Graph theory	Slides 11-19
3.3-3.7	Nodal voltage analysis	Slides 21-52
3.8-3.9	Mesh current analysis	Slides 54-67
3.10	Comparison of Nodal and mesh analysis	Slides 79-75
3.11	Application: DC transistor circuits	Slides 77-82
3.12	Summary and assignments	Slides 84-86

3.8 Mesh current analysis (a)

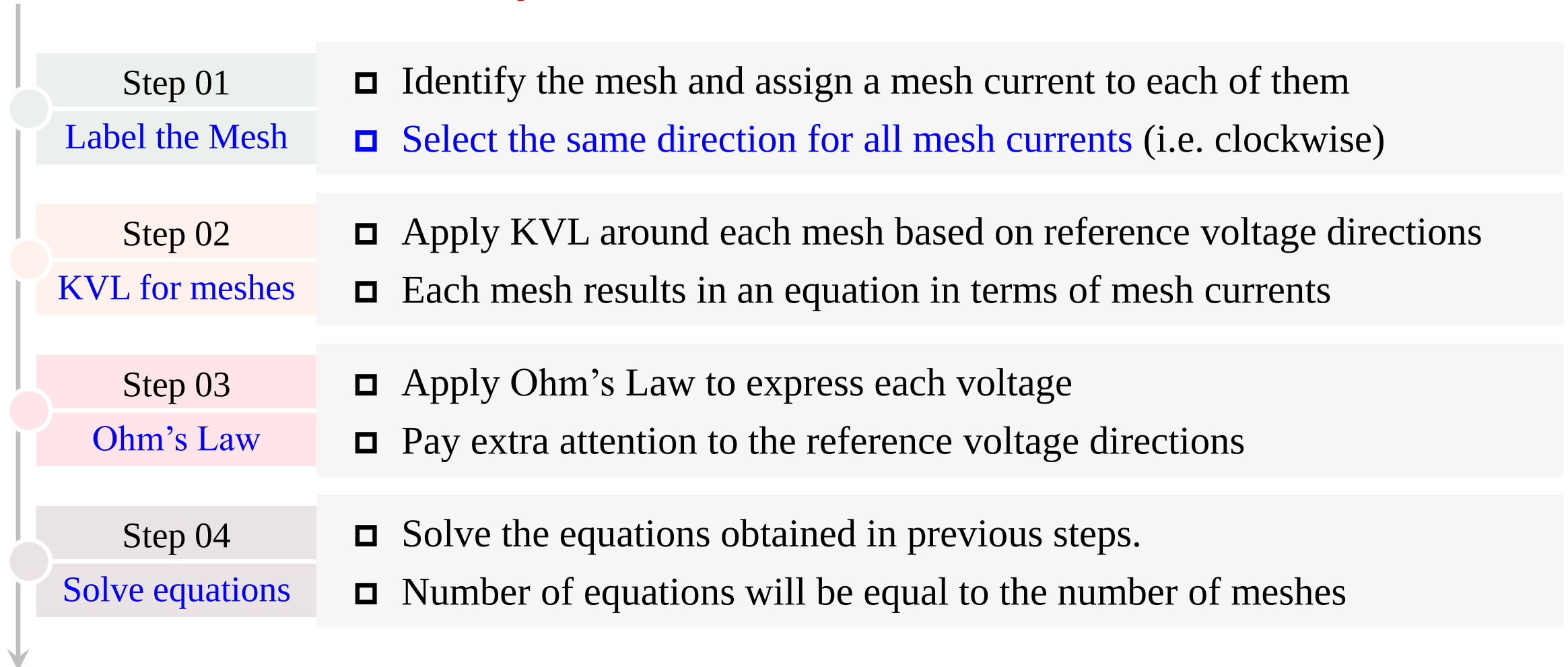
- Mesh current analysis

- a) The current flowing around a mesh is called mesh current (e.g., i_1 , i_2 in figure below).
- b) Mesh analysis (or the mesh current method) is a method that is used to solve planar circuits for the currents (and indirectly the voltages) at any place in the electrical circuit.
- c) Planar circuits are circuits that can be drawn on a plane surface with no wires crossing each other.
- d) Mesh analysis choose mesh current as the independent variables, and applied KVL along the meshes to establish the equations.



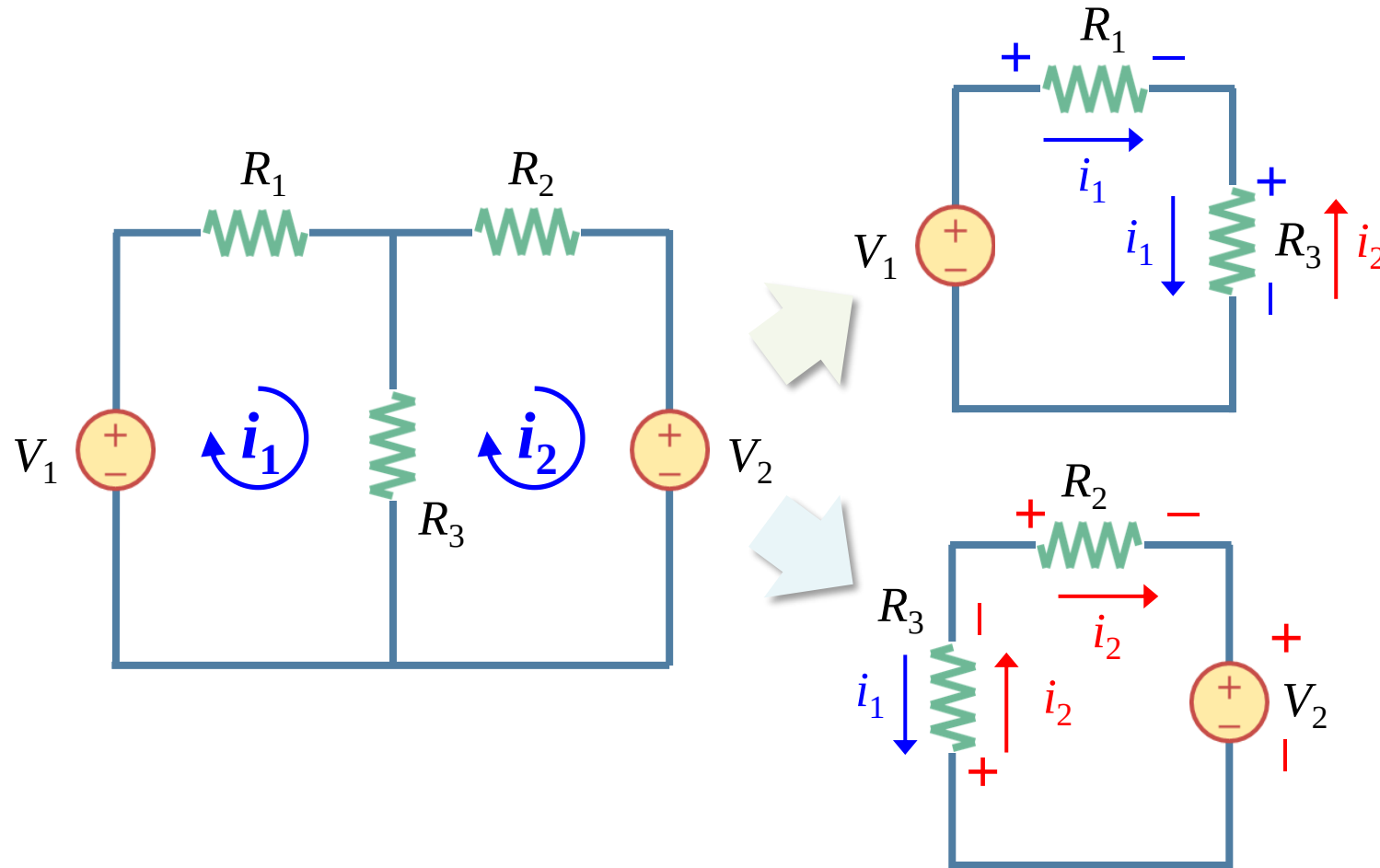
3.8 Mesh current analysis (b)

- Mesh current analysis



3.8 Mesh current analysis (c)

- Two-mesh circuits



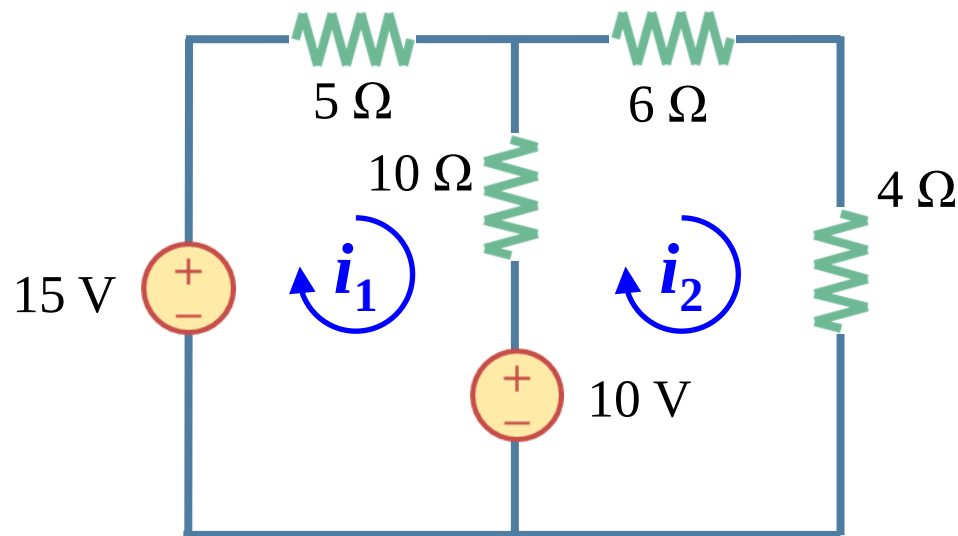
3.8 Mesh current analysis (d)

- An example

Step 01

Label the Mesh

- Identify the mesh and assign a mesh current to each of them
- Better to select the same direction for all mesh currents (i.e. clockwise)



(a) Find the mesh and label them with label currents.

Example 09: find the current in each branch.

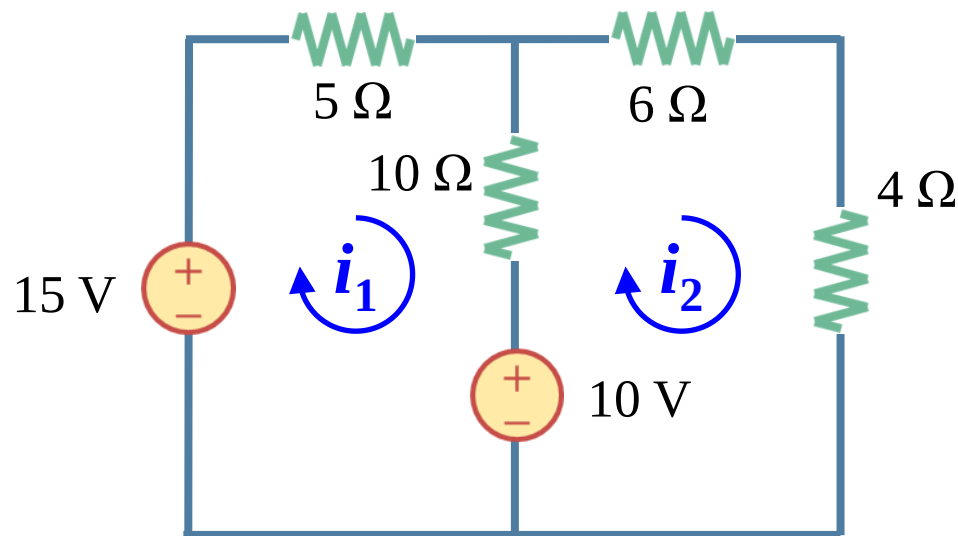
3.8 Mesh current analysis (e)

- An example

Step 02

KVL for meshes

- Apply KVL around each mesh based on reference voltage directions
- Each mesh results in an equation in terms of mesh currents



(a) Find the mesh and label them with label currents.

(b) Write KVL equations.

$$\begin{cases} -15 + v_1 + v_2 + 10 = 0 \\ -10 + v_2 + v_3 + v_4 = 0 \end{cases}$$

Example 09: find the current in each branch.

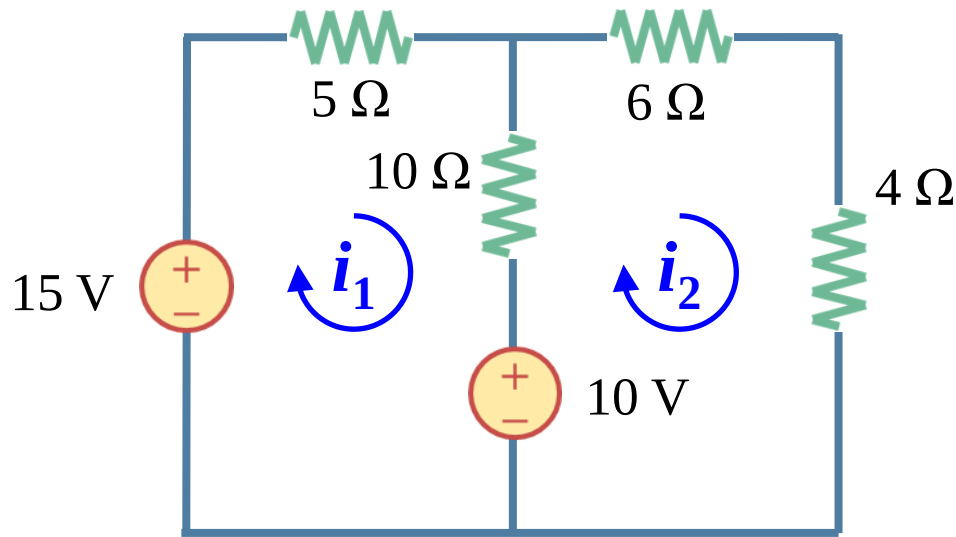
3.8 Mesh current analysis (f)

• An example

Step 03

Ohm's Law

- ❑ Apply Ohm's Law to express each voltage
- ❑ Pay extra attention to the reference voltage directions



Example 09: find the current in each branch.

(c) Rewrite the KVL equations in term of mesh currents using Ohm's Law

$$\begin{cases} -15 + v_1 + v_2 + 10 = 0 \\ -10 + v_2 + v_3 + v_4 = 0 \end{cases}$$



$$\begin{cases} -15 + 5i_1 + 10(i_1 - i_2) + 10 = 0 \\ -10 + 10(i_2 - i_1) + 6i_2 + 4i_2 = 0 \end{cases}$$

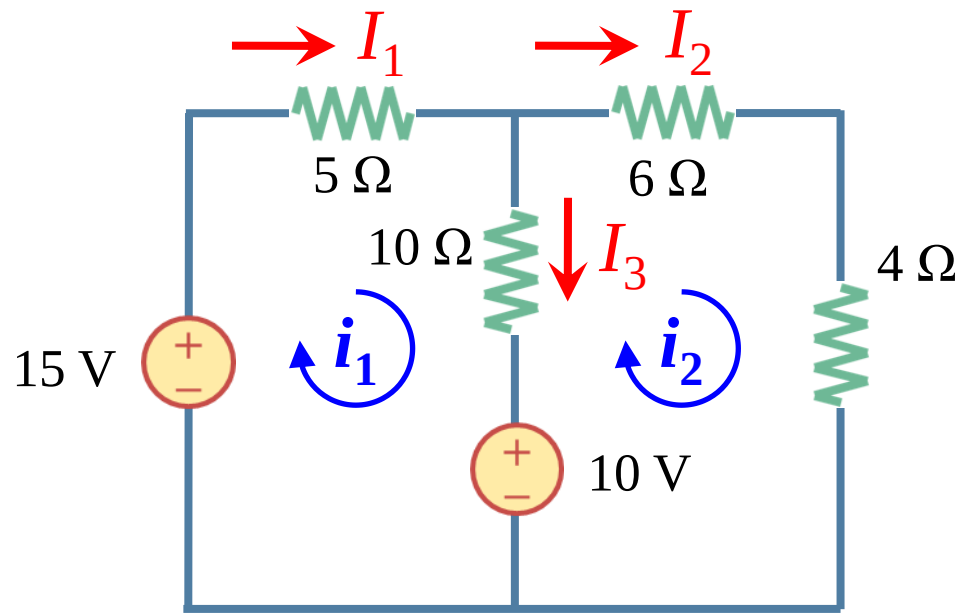
3.8 Mesh current analysis (g)

- An example

Step 04

Solve equations

- Solve the equations obtained in previous steps.
- Number of equations will be equal to the number of meshes



$$\begin{cases} -15 + 4i_1 + 6(i_1 - i_2) + 10 = 0 \\ -10 + 6(i_2 - i_1) + 6i_2 + 4i_2 = 0 \end{cases}$$

$$i_1 = i_2 = 1 \text{ A}$$

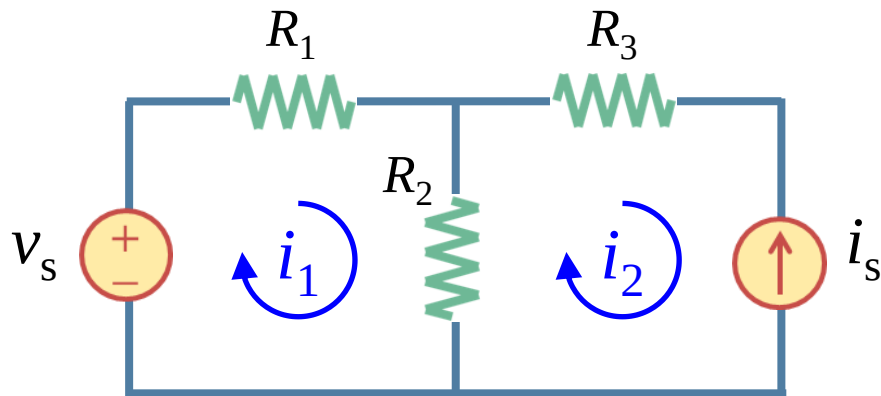
$$I_1 = i_1 = 1 \text{ A}, \quad I_2 = i_2 = 1 \text{ A}, \quad I_3 = i_1 - i_2 = 0 \text{ A}$$

Example 09: find the current in each branch.

3.9 Supermesh (a)

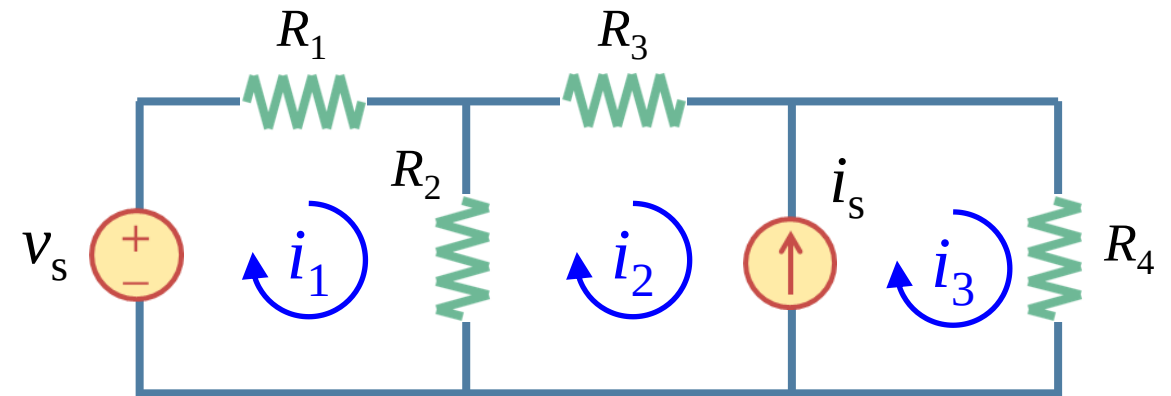
- Circuits with current sources

Case 1: the current source is not shared by meshes



$$\begin{cases} \text{Mesh 1: } -v_s + R_1 i_1 + R_2(i_1 - i_2) = 0 \\ \text{Mesh 2: } i_2 = -i_s \end{cases}$$

Case 2: the current source is shared by two meshes

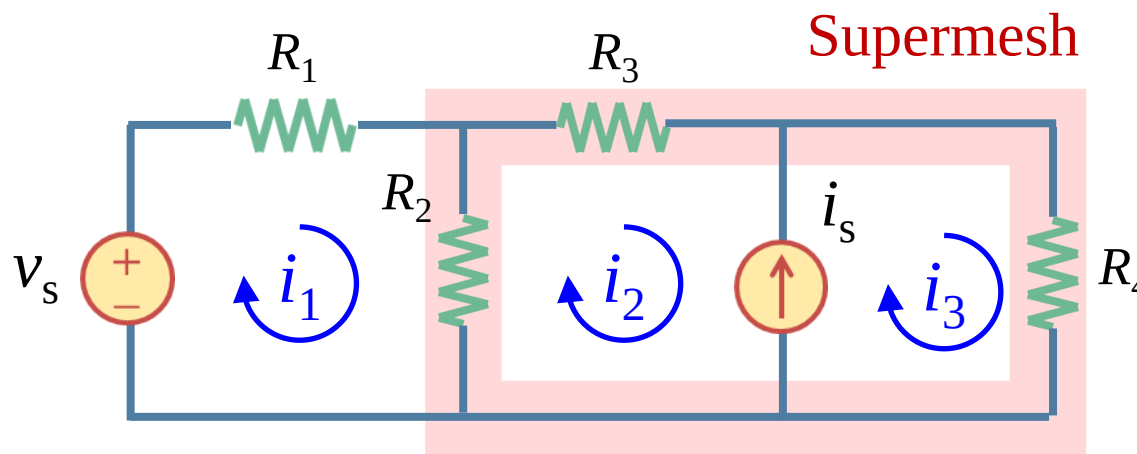


? i_2 & i_3 ?

3.9 Supermesh (b)

- Supermesh

- a) It can be difficult to apply mesh analysis when **two meshes share a common current source**, either dependent or independent, as the voltage drop at such sources is unknown and cannot be expressed with mesh current.
- b) To address this, the concept of **supermesh** (超网孔) are created, which actually reflects the fact that KVL can be applied for a loop in addition to a mesh.



1. Mesh 1: $-v_s + R_1 i_1 + R_2(i_1 - i_2) = 0$

2. Supermesh:

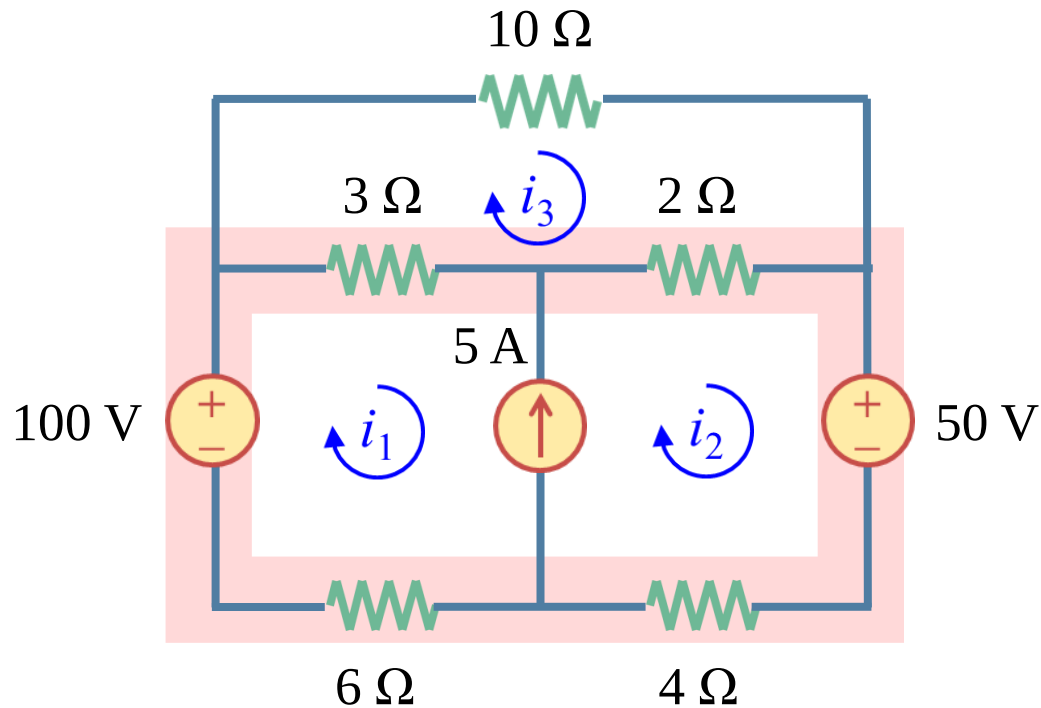
$$(i_2 - i_1)R_2 + i_2 R_3 + i_3 R_4 = 0$$

3. The shared current source:

$$i_3 - i_2 = i_s$$

3.9 Supermesh (c)

Example 09: Find i_1 , i_2 , i_3 in the following circuits.



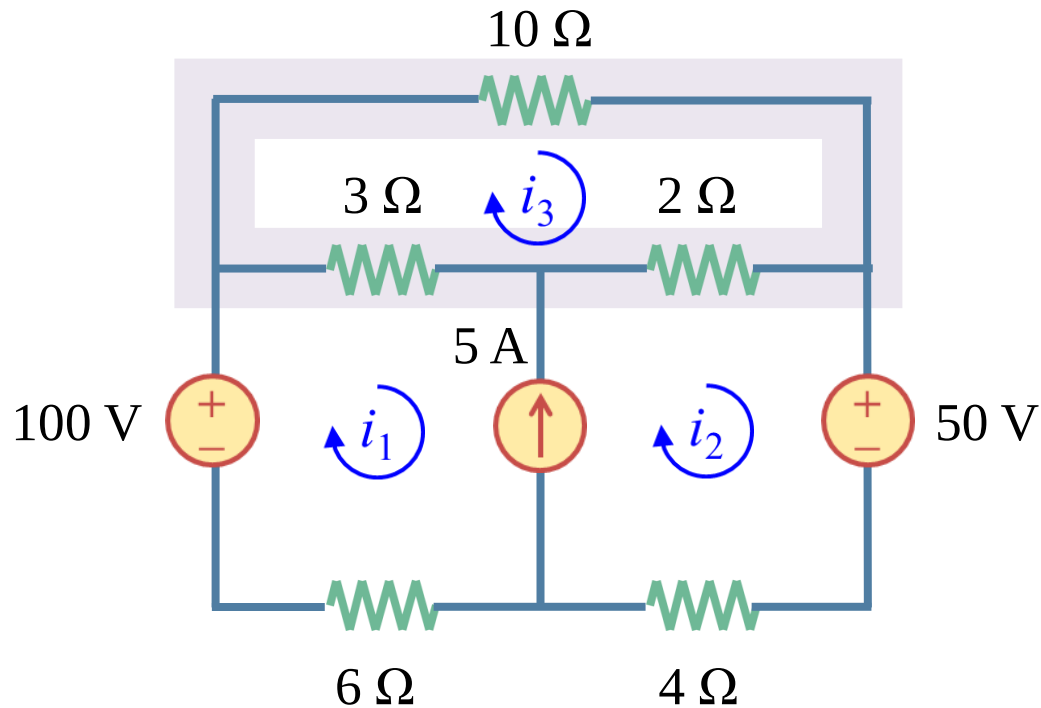
Solution:

(a) Label the supermesh and write the KVL equation:

$$-100 + 3(i_1 - i_3) + 2(i_2 - i_3) + 50 + 4i_2 + 6i_1 = 0$$

3.9 Supermesh (d)

Example 09: Find i_1 , i_2 , i_3 in the following circuits.



Solution:

(a) Label the supermesh and write the KVL equation:

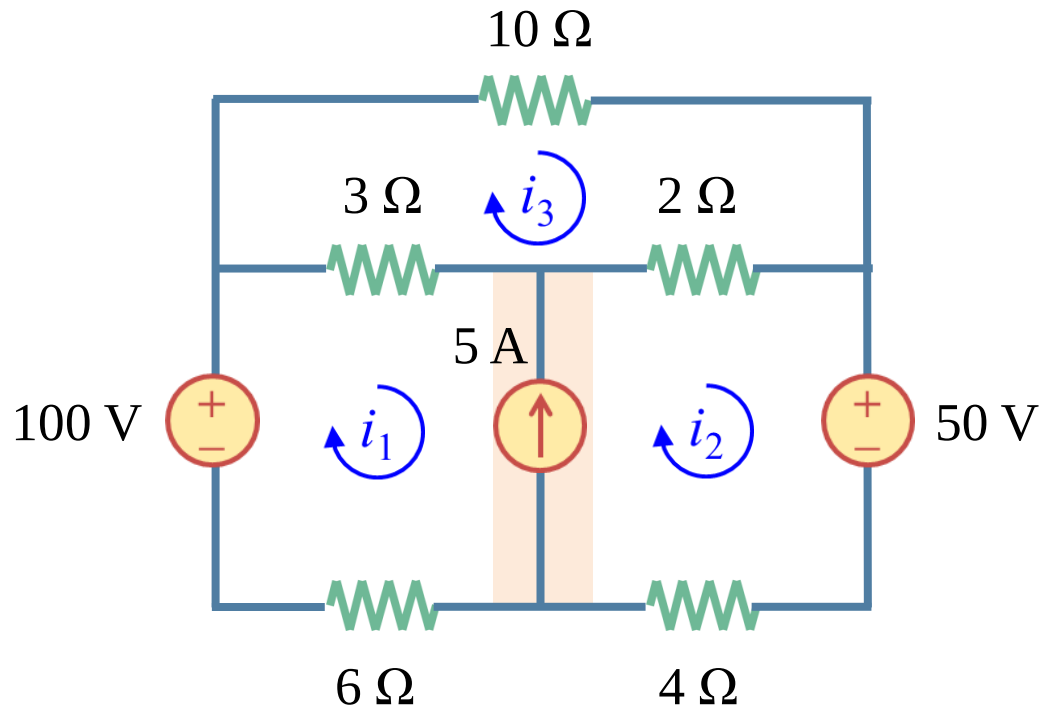
$$-100 + 3(i_1 - i_3) + 6i_1 + 2(i_2 - i_3) + 50 + 4i_2 = 0$$

(b) Write another KVL equation from mesh 3:

$$3(i_3 - i_1) + 10i_3 + 2(i_3 - i_2) = 0$$

3.9 Supermesh (e)

Example 09: Find i_1 , i_2 , i_3 in the following circuits.



Solution:

(a) Label the supermesh and write the KVL equation:

$$-100 + 3(i_1 - i_3) + 6i_1 + 2(i_2 - i_3) + 50 + 4i_2 = 0$$

(b) Write another KVL equation from mesh 3:

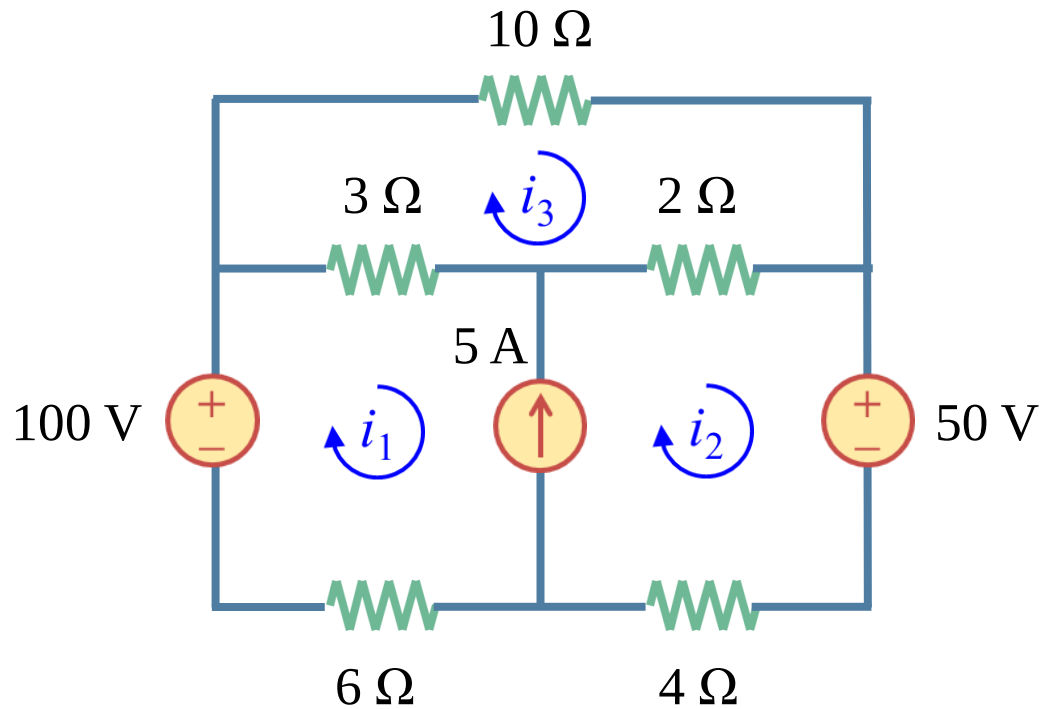
$$3(i_3 - i_1) + 10i_3 + 2(i_3 - i_2) = 0$$

(b) One more continuity equation from the branch with the shared current source:

$$i_2 - i_1 = 5$$

3.9 Supermesh (f)

Example 09: Find i_1 , i_2 , i_3 in the following circuits.



Solution:

(a) Now we have three equations:

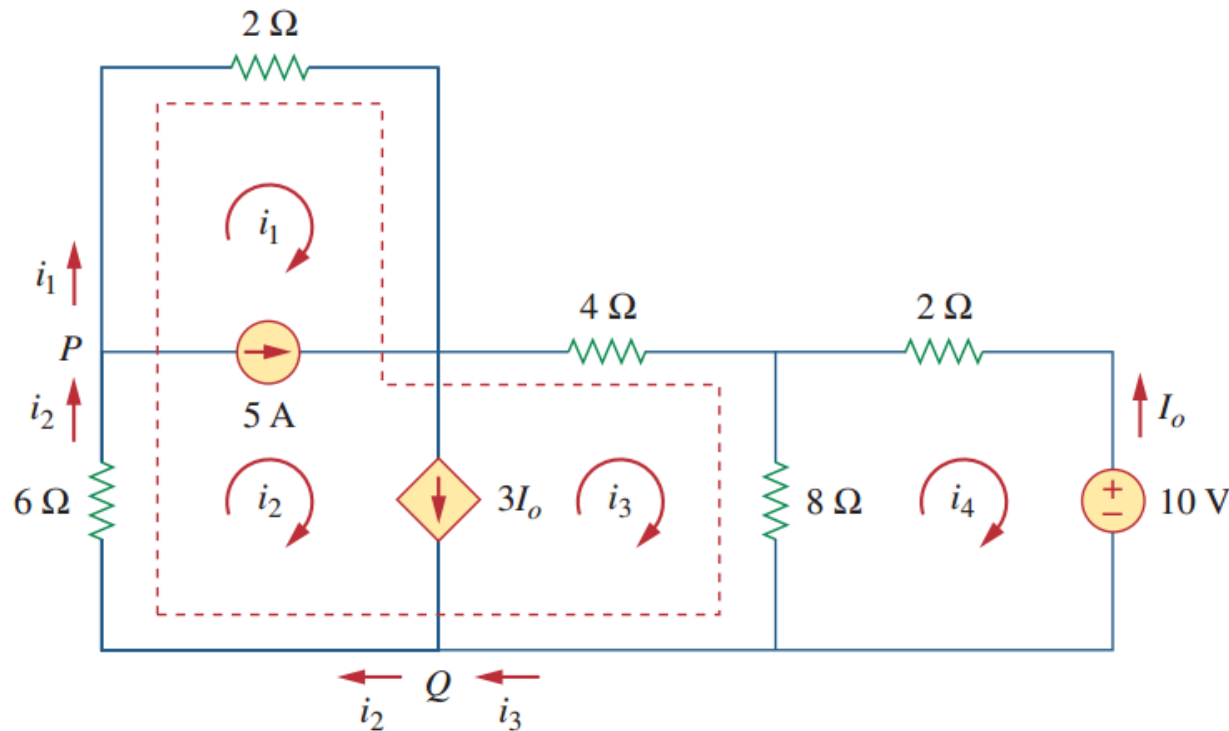
$$\begin{cases} -100 + 3(i_1 - i_3) + 6i_1 + 2(i_2 - i_3) + 50 + 4i_2 = 0 \\ 3(i_3 - i_1) + 10i_3 + 2(i_3 - i_2) = 0 \\ i_2 - i_1 = 5 \end{cases}$$



$$i_1 = 1.75 \text{ A} \quad i_2 = 6.75 \text{ A} \quad i_3 = 1.25 \text{ A}$$

3.9 Supermesh (g)

Example 10: Find i_1 , i_2 , i_3 , i_4 in the following circuits.



Solution:

(a) Supermesh:

$$2i_1 + 4i_3 + 8(i_3 - i_4) + 6i_2 = 0$$

(b) Mesh 4: $2i_4 + 8(i_4 - i_3) + 10 = 0$

(c) Continuity equations:

$$i_2 - i_1 = 5 \quad i_2 - i_3 = 3I_o$$

(d) Tanking account of:

$$I_o = -i_4$$

$$i_1 = -7.5 \text{ A}, i_2 = -2.5 \text{ A}, i_3 = 3.93 \text{ A}, i_4 = 2.143 \text{ A}$$

Lesson 3 Systematic approaches for circuit analysis

3.1-3.2	Graph theory	Slides 11-19
3.3-3.7	Nodal voltage analysis	Slides 21-52
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3.10	Comparison of Nodal and mesh analysis	Slides 79-75
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3.12	Summary and assignments	Slides 84-86

3.10 Comparison of the two methods (a)

- Comparison between nodal and mesh analysis

Method	To identify	Law	In term of	Can be applied to
Nodal voltage analysis	nodes (+ supernodes)	KCL	node voltages	all circuits
Mesh current analysis	meshes (+ supermesh)	KVL	mesh currents	planar circuits

3.10 Comparison of the two methods (b)

- How to select between nodal and mesh analysis?

The nature of the circuit + the information desired

Nodal analysis is good for

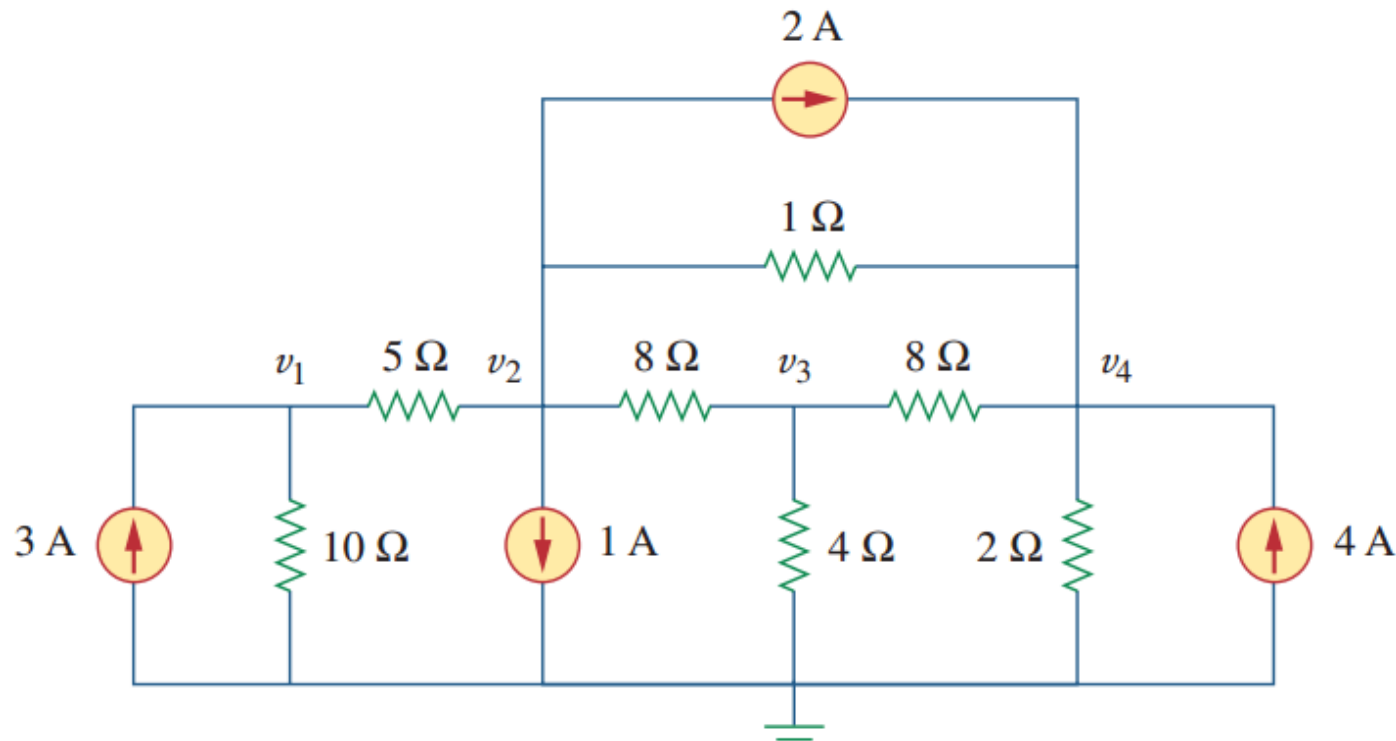
- The network contains many elements connected in parallel.
- The network contains many current sources.
- The network contains many supermeshes.
- The circuit has fewer nodes than meshes (节点数少于网孔数)
- Node voltages are what being solved for

Mesh analysis is good for

- The network contains many serially connected elements.
- The network contains many voltage sources.
- The network contains many supernodes.
- The circuit has fewer meshes than nodes (网孔数少于节点数).
- Branch or mesh currents are what being solved for

3.10 Comparison of the two methods (c)

Example 11: Find the nodal voltages in the following circuits.



Solution:

(a) Select nodal analysis because:

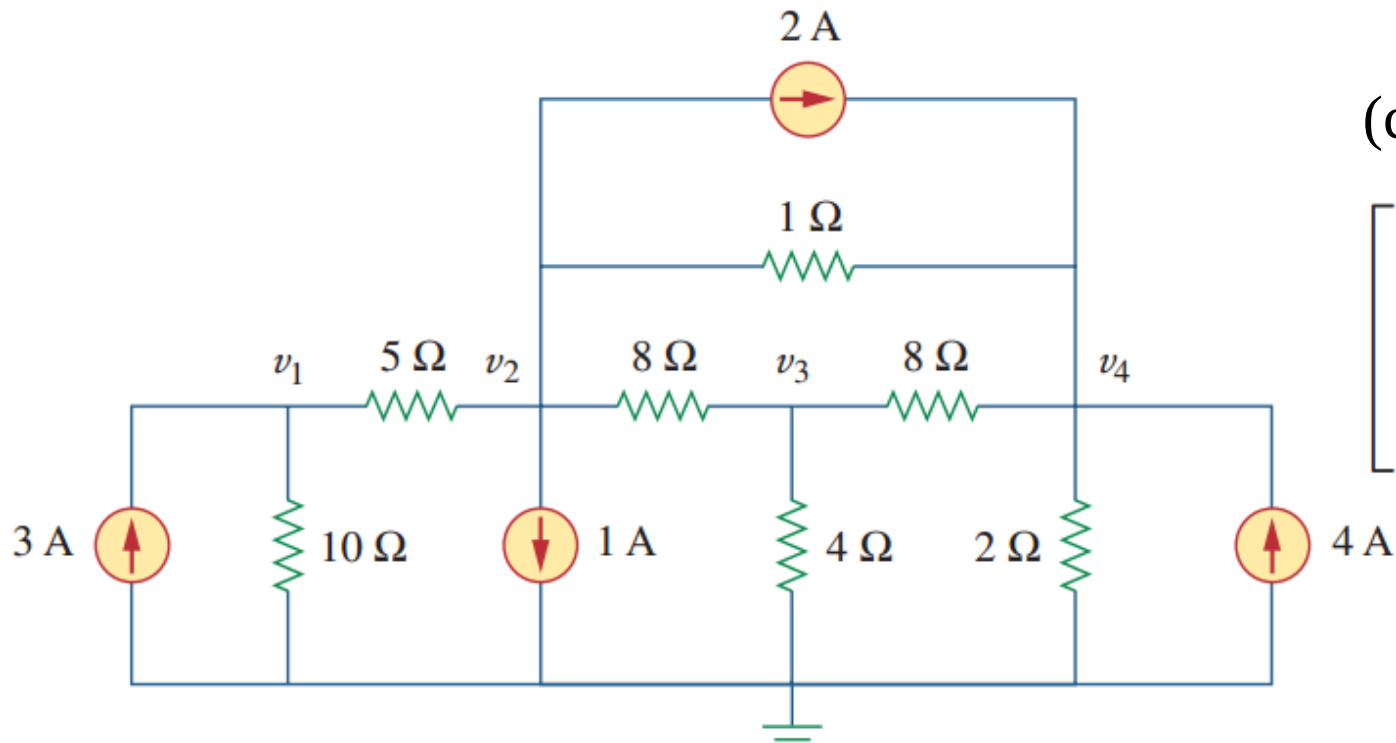
Many current sources, many parallel connected elements, and the voltages are what wanted.

(b) Follow the steps of nodal analysis:

$$\begin{cases} v_1/10 + (v_1 - v_2)/5 = 3 \\ (v_2 - v_1)/5 + (v_2 - v_4)/1 + (v_2 - v_3)/8 = -2 - 1 \\ (v_3 - v_2)/8 + (v_3 - v_4)/8 + v_3/4 = 0 \\ (v_4 - v_2)/1 + (v_4 - v_3)/8 + v_4/2 = 2 + 4 \end{cases}$$

3.10 Comparison of the two methods (d)

Example 11: Find the nodal voltages in the following circuits.



Solution:

(c) Simplifying and resolving the equations

$$\begin{bmatrix} 0.3 & -0.2 & 0 & 0 \\ -0.2 & 1.325 & -0.125 & -1 \\ 0 & -0.125 & 0.5 & -0.125 \\ 0 & -1 & -0.125 & 1.625 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = \begin{bmatrix} 3 \\ -3 \\ 0 \\ 6 \end{bmatrix}$$

$$v_1 = 13.897 \text{ V}$$

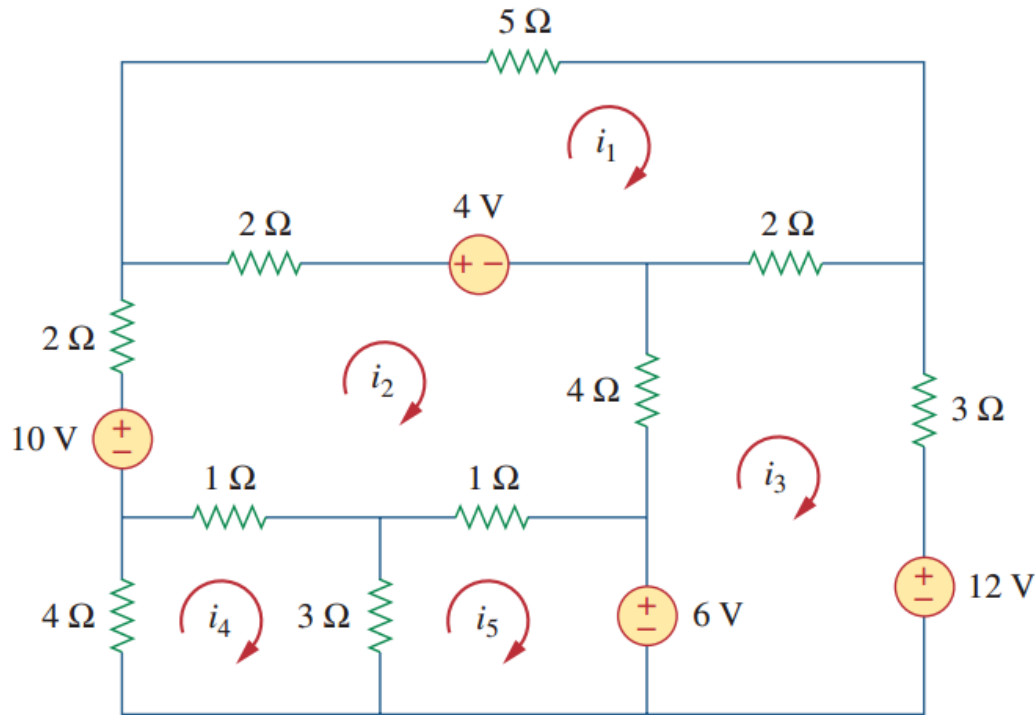
$$v_2 = 5.845 \text{ V}$$

$$v_3 = 3.348 \text{ V}$$

$$v_4 = 7.547 \text{ V}$$

3.10 Comparison of the two methods (e)

Example 12: Find the mesh currents in the following circuits.



Solution:

(a) Select mesh analysis because:
many voltage sources, many serially connected elements, and the currents are what wanted.

(b) Follow the steps for mesh analysis:

$$-4 + 2(i_1 - i_2) + 5i_1 + 2(i_1 - i_3) = 0$$

$$-10 + 2i_2 + 2(i_2 - i_1) + 4 + 4(i_2 - i_3) + 1(i_2 - i_5) + 1(i_2 - i_4) = 0$$

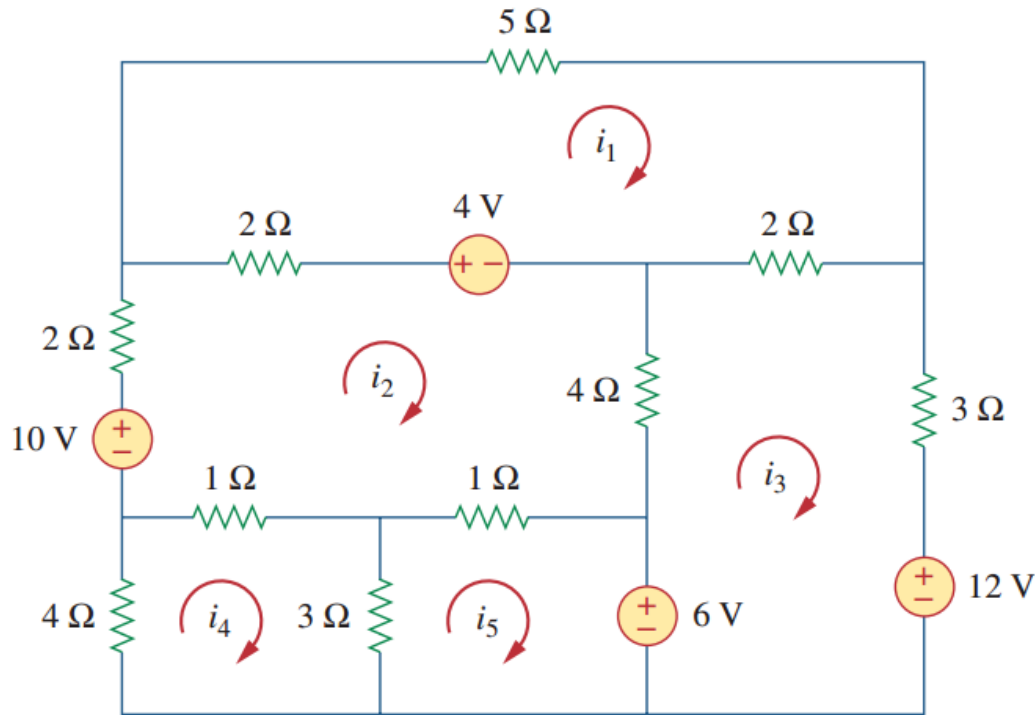
$$-6 + 4(i_3 - i_2) + 2(i_3 - i_1) + 3i_3 + 12 = 0$$

$$4i_4 + 1(i_4 - i_2) + 3(i_4 - i_5) = 0$$

$$3(i_5 - i_4) + 1(i_5 - i_2) + 6 = 0$$

3.10 Comparison of the two methods (f)

Example 12: Find the mesh currents in the following circuits.



Solution:

(c) Simplifying and resolving the equations

$$\begin{bmatrix} 9 & -2 & -2 & 0 & 0 \\ -2 & 10 & -4 & -1 & -1 \\ -2 & -4 & 9 & 0 & 0 \\ 0 & -1 & 0 & 8 & -3 \\ 0 & -1 & 0 & -3 & 4 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \\ -6 \\ 0 \\ -6 \end{bmatrix}$$

$\Rightarrow$

$$\begin{aligned} i_1 &= 0.3829\ \text{A}; i_2 = 0.2110\ \text{A}; \\ i_3 &= -0.4878\ \text{A}; i_4 = -0.7184\ \text{A}; \\ i_5 &= -1.9860\ \text{A} \end{aligned}$$

Supplementary: Cramer's Rule (克拉默法则)

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \cdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \cdots \\ b_n \end{bmatrix}$$

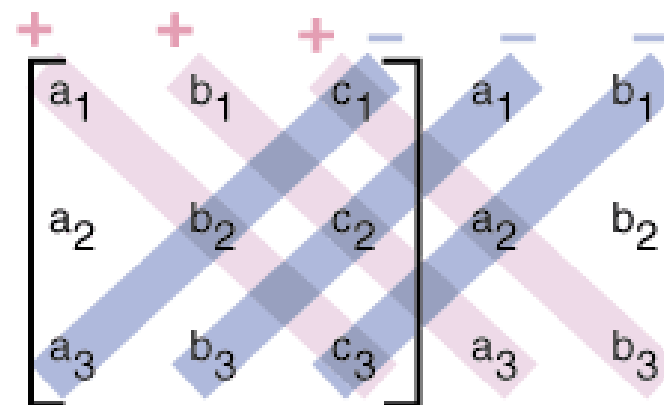
The solution of linear simultaneous equations by Cramer's rule boils down to finding

$$x_k = \frac{\Delta_k}{\Delta}, \quad k = 1, 2, \dots, n \quad (\text{A.12})$$

where Δ is the determinant of matrix $\mathbf{A}$ and Δ_k is the determinant of the matrix formed by replacing the k th column of $\mathbf{A}$ by $\mathbf{B}$.

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}, \quad \Delta_1 = \begin{vmatrix} b_1 & a_{12} & \cdots & a_{1n} \\ b_2 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ b_n & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$

$$\Delta_2 = \begin{vmatrix} a_{11} & b_1 & \cdots & a_{1n} \\ a_{21} & b_2 & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1} & b_n & \cdots & a_{nn} \end{vmatrix}, \dots, \Delta_n = \begin{vmatrix} a_{11} & a_{12} & \cdots & b_1 \\ a_{21} & a_{22} & \cdots & b_2 \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1} & a_{n2} & \cdots & b_n \end{vmatrix}$$



$$\det \mathbf{A} = (a_1 b_2 c_3 + b_1 c_2 a_3 + c_1 a_2 b_3) - (a_3 b_2 c_1 + b_3 c_2 a_1 + c_3 a_2 b_1)$$

Lesson 3 Systematic approaches for circuit analysis

3.1-3.2	Graph theory	Slides 11-19
3.3-3.7	Nodal voltage analysis	Slides 21-52
3.8-3.9	Mesh current analysis	Slides 54-67
3.10	Comparison of Nodal and mesh analysis	Slides 79-75
3.11	Application: DC transistor circuits	Slides 77-82
3.12	Summary and assignments	Slides 84-86

3.11 Application: DC transistor circuits (a)

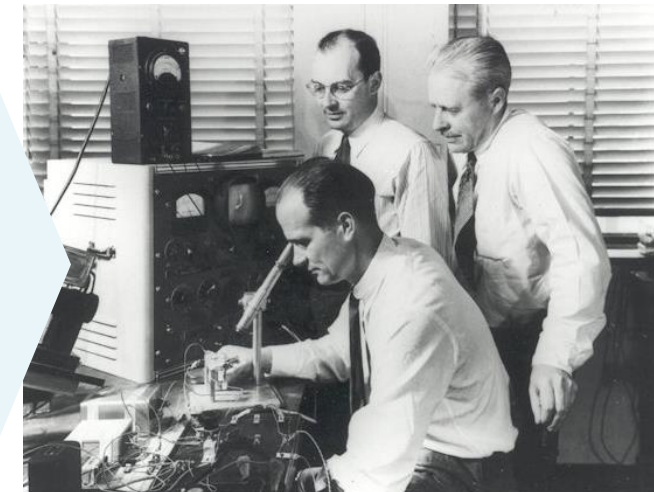
- Transistors

- a) A transistor is a semiconductor device used to amplify or switch electronic signals and electrical power.
- b) Transistors are one of the basic building blocks of modern electronics. It is composed of semiconductor material usually with at least three terminals for connection to an external circuit.
- c) There are many types of transistors, including bipolar junction transistors, field-effect transistors, tunneling transistors, phototransistors, etc.



Replica of the first transistor demonstrated in 1947/12/23.

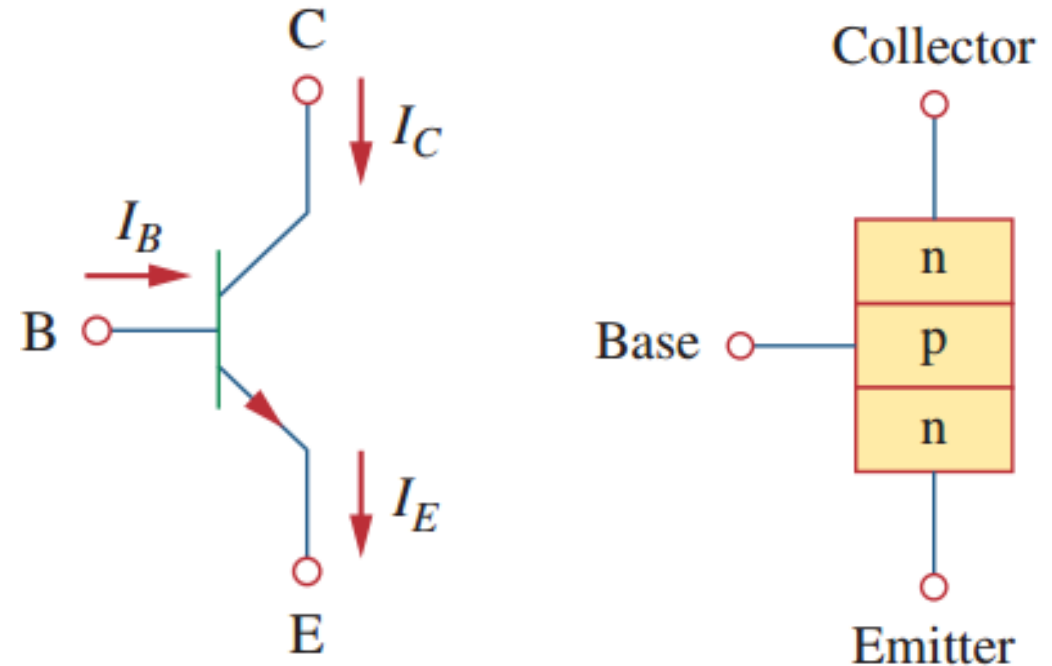
Transistor inventors, John Bardeen, and Walter Brattain, with their supervisor William Shockley (seated) 1948. (Credit: AT&T Archives.)



3.11 Application: DC transistor circuits (b)

- Bipolar junction transistors (BJTs)

- a) A bipolar junction transistor (BJT) is a type of transistor that uses both electrons and holes as charge carriers, which is different from unipolar transistors.
- b) A BJT can be based on both NPN or PNP.
- c) BJT has so many applications. For instance, many power amplifiers in RF fronts of cell phones are made by BJTs, thanks their high efficiency and high linearity.

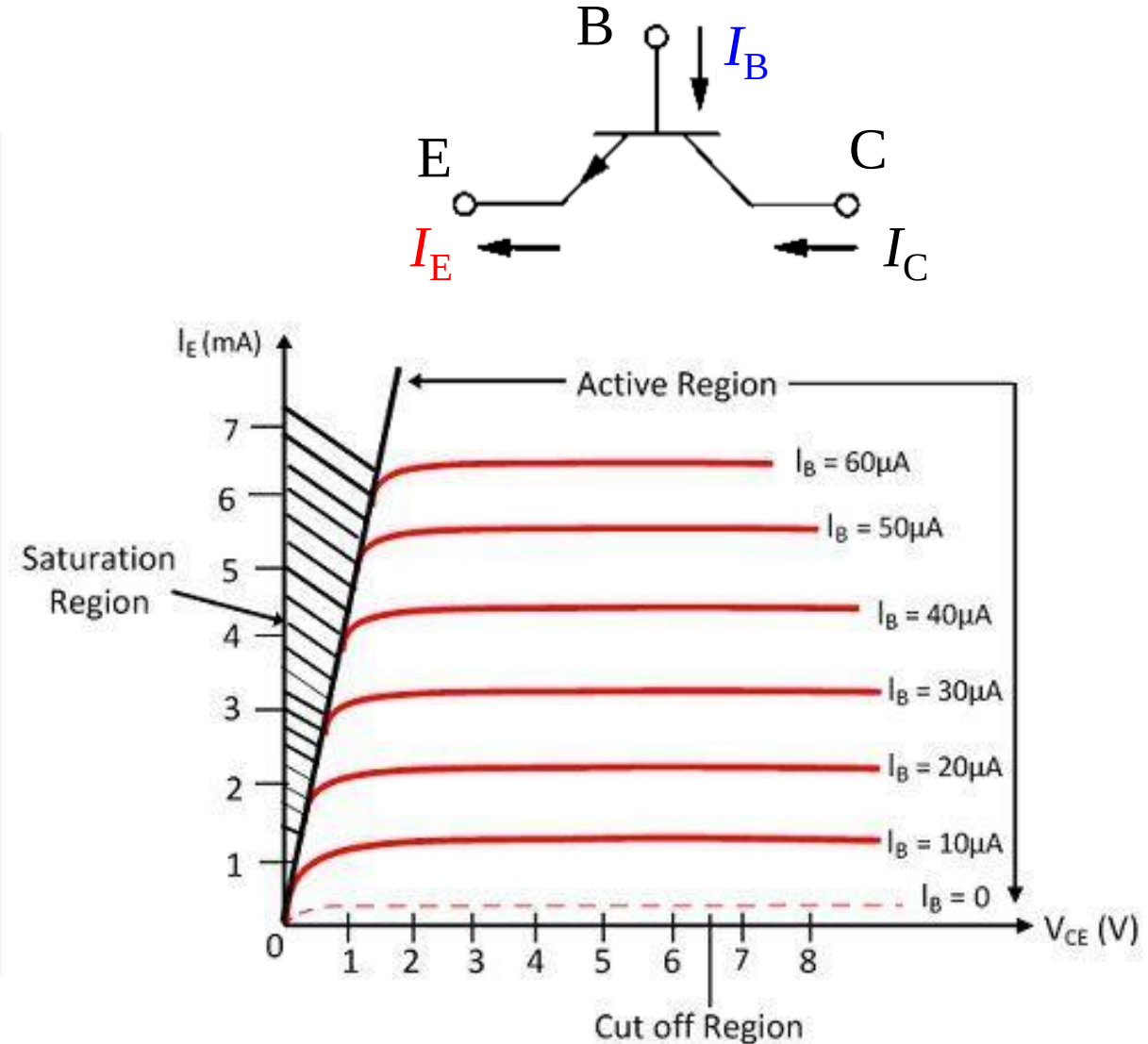


$$I_E = I_B + I_C, I_E = (1 + \beta) \cdot I_B$$

3.11 Application: DC transistor circuits (c)

- I - V characteristics of a BJT

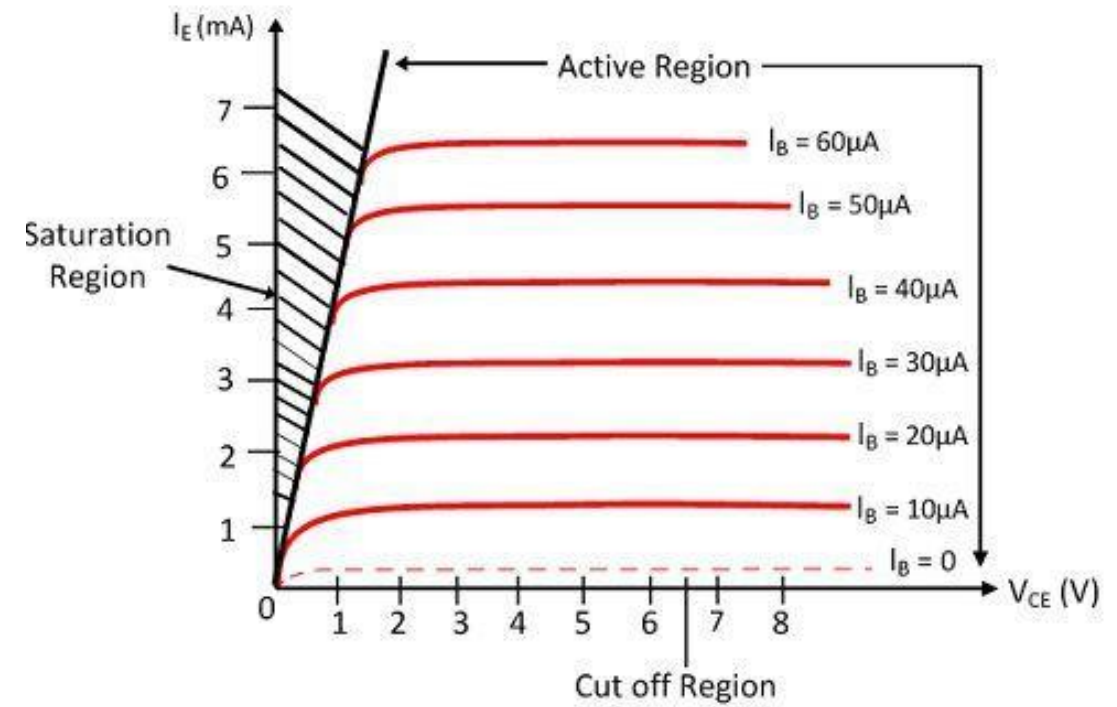
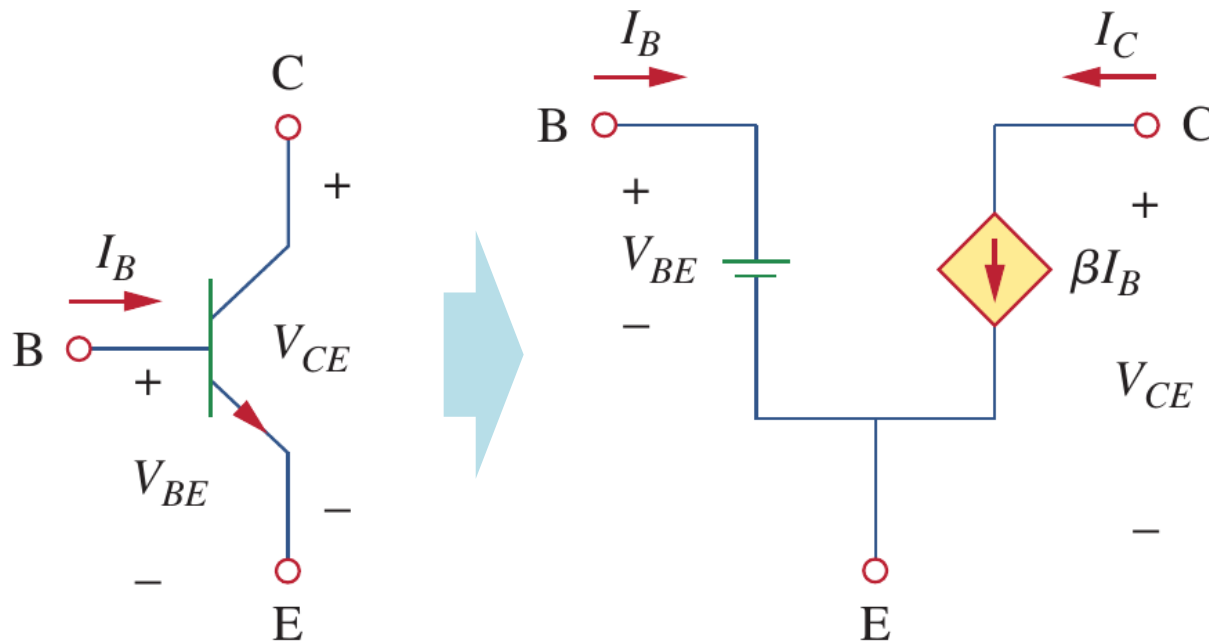
- Three operating modes: 1) active/放大, 2) cutoff/截止, and 3) saturation/饱和. The active mode (放大模式) is the most commonly used one for analog circuit.
- In active mode, $V_{BE} \approx 0.7$ V (silicon BJTs).
- The currents I_C and I_B are related by $I_C = \beta I_B$, where β is called the common-emitter current gain, taking a value from 50 to 1000.
- The currents I_C and I_E are related by $I_C = \alpha I_E$, where α is called the common-base current gain, taking a value from 0.98 to 0.999.



3.11 Application: DC transistor circuits (d)

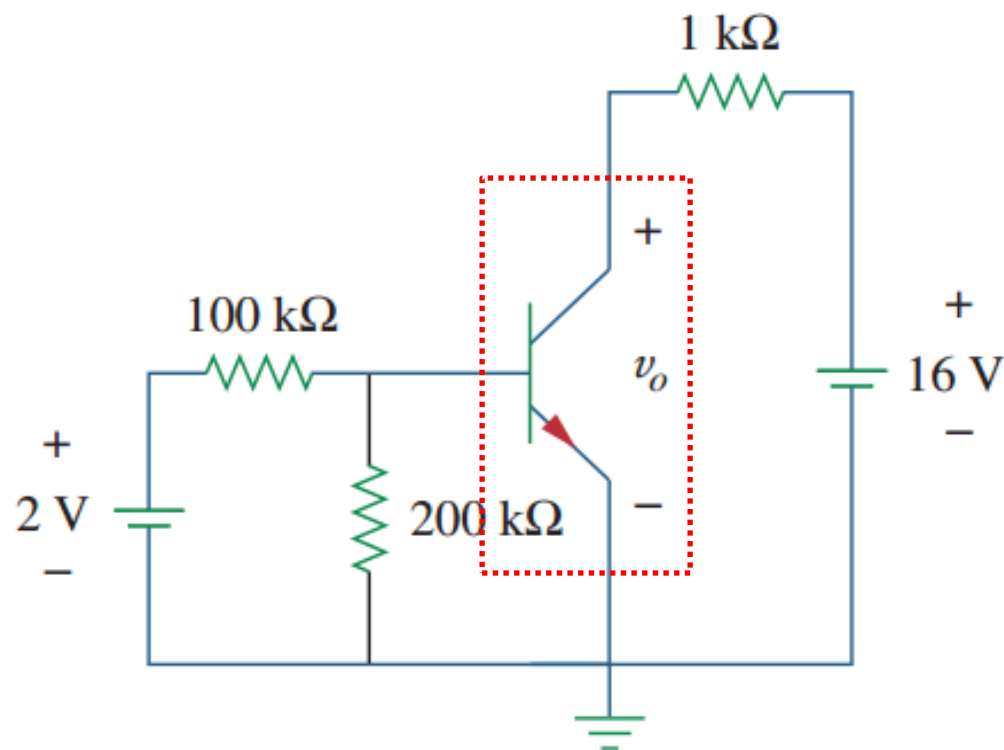
- Equivalent circuits of BJTs

In active mode, a BJT can be modeled as a dependent current-controlled current source



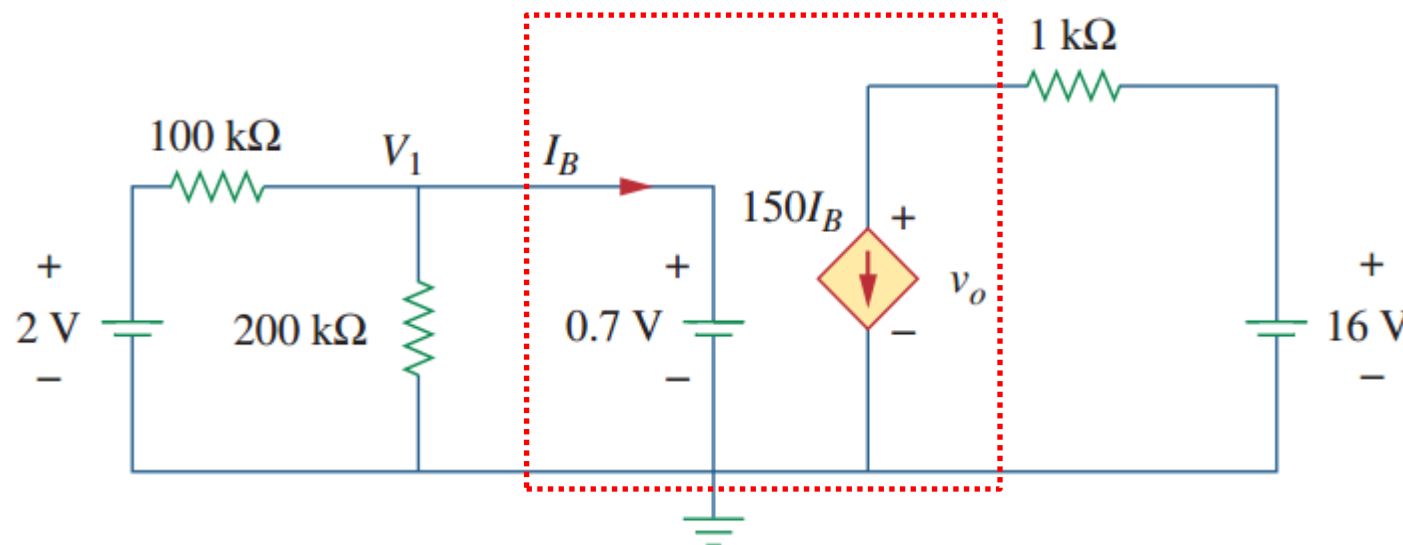
3.11 Application: DC transistor circuits (e)

Example 13: For the BJT circuit in Fig. 3.43, $\beta = 150$ and $V_{BE} = 0.7$ V. Find v_o .



Solution:

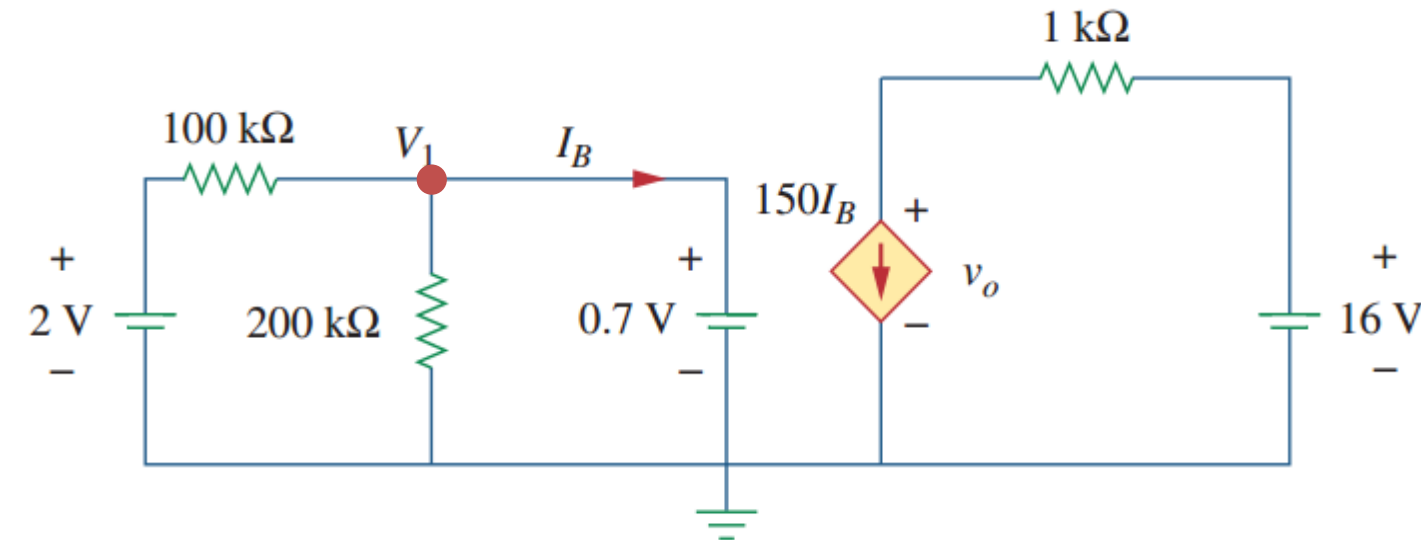
(a) Replace the transistor with its equivalent circuit:



3.11 Application: DC transistor circuits (f)

Example 13: For the BJT circuit in Fig. 3.43, $\beta = 150$ and $V_{BE} = 0.7$ V. Find v_o .

Solution:



(b) nodal voltage analysis:

– KCL at node 1:

$$(0.7 - 2)/100k + 0.7/200k + I_B = 0$$

– Expression of the dependent source

$$I_C = 150I_B = (16 - v_o)/1k$$

(c) Solve the equations:

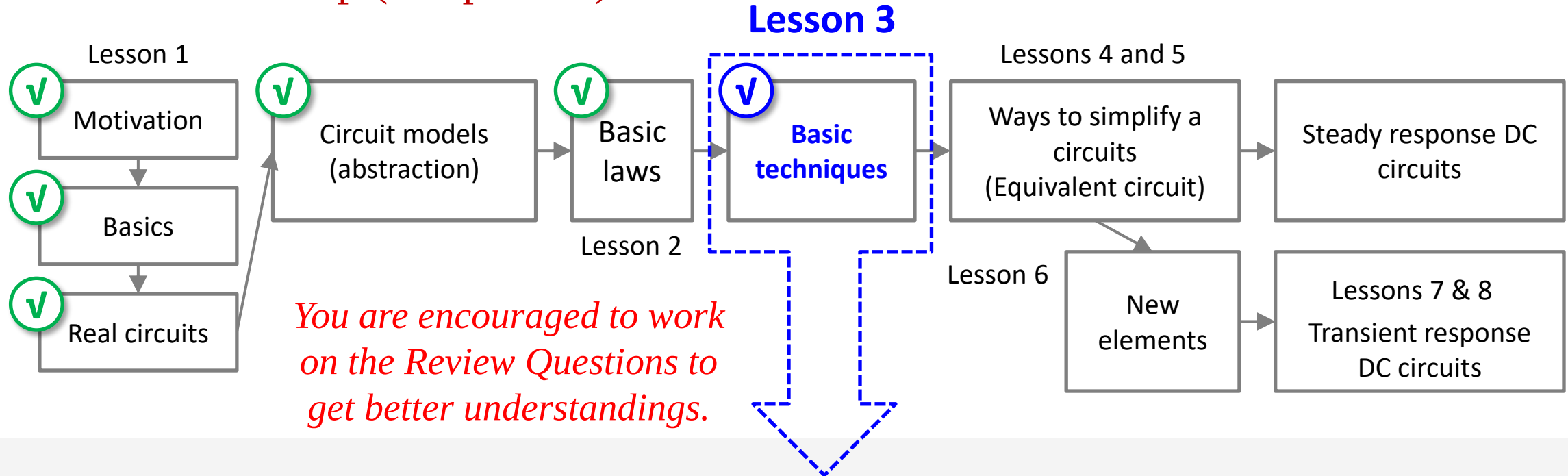
$$v_o = 14.575 \text{ V}$$

Lesson 3 Systematic approaches for circuit analysis

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3.12 Summary and assignments (a)

- EE104 roadmap (DC portion)



Nodal analysis & Mesh analysis

(planar circuits & supernode & supermesh)

3.12 Summary and assignments (b)

- **Nodal analysis** is the application of Kirchhoff's current law (KCL) at the non-reference nodes. We express the result in terms of the node voltages. Solving the simultaneous equations yields the node voltages.
- **A supernode** consists of two non-reference nodes connected by a (dependent or independent) voltage source.
- **Mesh analysis** is the application of Kirchhoff's voltage law (KVL) around meshes, i.e., loops without other loops inside them. We express the result in terms of mesh currents. Solving the simultaneous equations yields the mesh currents.
- **A supermesh** consists of two meshes that have a (dependent or independent) current source in common.
- Nodal analysis is normally used when a circuit has fewer node equations than mesh equations. Mesh analysis is normally used when a circuit has fewer mesh equations than node equations.
- DC transistor circuits can be analyzed using the techniques covered in this chapter.

3.12 Summary and assignments (c)

Chapter 3:

Problems 3.8, 3.23, 3.40, 3.45, 3.62, 3.91

- (a) A certain amount of practicing is necessary to sharpen your mind and to deepen your understanding.
- (b) There are plenty of problems in the textbook, try to complete as many as you can.

Important updates:

- DC circuits:

		Class 1	Class 4	Assignments	Assignments to hand in
Lesson 01	Basic Concepts	09/10	09/10	No	--
Lesson 02	Basic Laws	09/24	09/24	Yes	--
Lesson 03	Methods of Analysis	10/08	10/08	Yes	Assignments of Lesson 02
Lesson 04	Circuit Theorems	10/12	10/12	Yes	--
Lesson 05	Operational Amplifiers	10/15	10/15	Yes	Assignments of Lesson 03
Lesson 06	Capacitors and Inductors	10/22	10/22	Yes	Assignments of Lesson 04
Lesson 07	First-Order Circuits	10/29	10/29	Yes	Assignments of Lesson 05
Lesson 08	Second-Order Circuits	11/05	11/05	No	Assignments of Lesson 06
Mid exam		To be announced			

Important updates:

- AC circuits:

		Class 1	Class 4	Assignments	Assignments to hand in
Lesson 09	Sinusoids and Phasors	11/12	11/12	Yes	Assignments of Lesson 07
Lesson 10	AC circuits	11/19	11/19	Yes	Assignments of Lesson 09
Lesson 11	Frequency Response	11/26	11/26	Yes	Assignments of Lesson 10
Lesson 12	Magnetically coupled circuits	12/03	12/03	Yes	Assignments of Lesson 11
Lesson 13	Laplace Transform	TBD	TBD	Yes	Assignments of Lesson 12
Lesson 14	Application of Laplace Transform	12/17	12/17	Yes	Assignments of Lesson 13
Lesson 15	Review	12/24	12/24		
Final exam		To be announced			--

Rocket clouds

